

Inhibition of the Notch signal transducer CSL by Pkc53E-mediated phosphorylation to fend off parasitic immune challenge in *Drosophila*

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Abstract

Notch signalling activity regulates hematopoiesis in *Drosophila* and vertebrates alike. Parasitoid wasp infestation of *Drosophila* larvae, however, requires a timely downregulation of Notch activity to allow the formation of encapsulation-active blood cells. Here we show that the *Drosophila* CSL transcription factor Suppressor of Hairless [Su(H)] is phosphorylated at Serine 269 in response to parasitoid wasp infestation. As this phosphorylation interferes with the DNA-binding of Su(H), it reversibly precludes its activity. Accordingly, phospho-deficient *Su(H)*^{S269A} mutants are immune compromised. A screen for kinases involved in Su(H) phosphorylation identified Pkc53E, required for normal hematopoiesis as well as for parasitoid immune response. Genetic and molecular interactions support the specificity of the Su(H)-Pkc53E relationship. Moreover, phorbol ester treatment inhibits Su(H) activity *in vivo* and in human cell culture. We conclude that Pkc53E targets Su(H) during parasitic wasp infestation, thereby remodeling the blood cell population required for wasp egg encapsulation.

eLife Assessment

This **valuable** study focuses on the regulation of Notch signaling during the immune response in *Drosophila*. The authors provide **solid** evidence in support of roles for Su(H) and Pkc53E-induced phosphorylation in *Drosophila* immunity. The work will be of interest to colleagues in immunity and receptor signaling.

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Introduction

Drosophila melanogaster harbours a sophisticated cellular immune system to fight invaders. The larval hematopoietic system comprises a circulating and sessile compartment of embryonic origin and the developing larval lymph gland, constituting a hematopoietic organ. The circulating and

sessile compartment consists primarily of macrophage-like plasmatocytes, plus a small number of crystal cells involved in wound healing and melanization responses to neutralize pathogens. Both cell types differentiate from hemocyte precursors within the lymph gland as well, and they are released during pupal stages to serve the adult with immune cells. Finally, lamellocytes represent the third blood cell type in *Drosophila*. While merely absent in healthy animals, their formation is induced within hours of wasp infestation or wounding (reviewed in: Banerjee et al., 2019 [\[1\]](#); Letourneau et al., 2016 [\[2\]](#); Hultmark and Andó, 2022 [\[3\]](#)). In fact, parasitism by endo-parasitoid wasps represents one of the most severe naturally occurring immune challenges to *Drosophila*, invariably causing death if not defended off properly. Parasitoid wasps deposit their egg into the live *Drosophila* larva, where it develops into the next wasp generation egressing from the pupa instead of a fly. Hence, wasp infestation is a life-threatening challenge to the *Drosophila* host, demanding an immediate immune response to overwhelm the parasite. This involves a massive increase in circulating hemocytes, and most energy resources are devoted to fight the invader. Here, lamellocytes play a critical role: they encapsulate the wasp egg and, by expressing Prophenoloxidase, induce a melanization reaction retarding further wasp development (Dudzić et al., 2015 [\[4\]](#); reviewed in: Banerjee et al., 2019 [\[1\]](#); Letourneau et al., 2016 [\[2\]](#); Hultmark and Andó, 2022 [\[3\]](#)).

Wasp infestation substantially remodels the composition of the *Drosophila* hemocyte population (Cattenoz et al., 2020 [\[5\]](#); Cho et al., 2020 [\[6\]](#); Tattikota et al., 2020 [\[7\]](#); reviewed in: Csordás et al., 2021 [\[8\]](#)). There is a vast increase in plasmatocytes and intermediate precursors in both hematopoietic compartments, from which lamellocyte differentiate. This process requires the combined regulatory input of several pathways, including JAK/STAT, JNK, Toll, Notch, EGFR and INR pathways, which in fact regulate blood cell homeostasis in general (reviewed in Banerjee et al., 2019 [\[1\]](#); Letourneau et al., 2018 [\[9\]](#); Csordás et al., 2021 [\[8\]](#)). Simultaneous to the massive expansion of lamellocytes, crystal cells are significantly reduced (Crozatier et al., 2004 [\[10\]](#); Krzemien et al., 2010 [\[11\]](#); Ferguson and Martinez-Agosto, 2014 [\[12\]](#); Cattenoz et al., 2020 [\[5\]](#); Cho et al., 2020 [\[6\]](#); Tattikota et al., 2020 [\[7\]](#); reviewed in Csordás et al., 2021 [\[8\]](#)). Crystal cells are generated both in the larval lymph gland as well as by trans-differentiation from plasmatocytes in the sessile compartment. Their formation, differentiation and survival strictly depend on Notch signalling activity (reviewed in: Banerjee et al., 2019 [\[1\]](#); Csordás et al., 2021 [\[8\]](#); Hultmark and Andó, 2022 [\[3\]](#)). Lamellocyte and crystal cell lineages are mutually exclusive. Accordingly, while promoting crystal cell fate, Notch activity inhibits differentiation of lamellocytes within the lymph gland, however, is reduced upon wasp infestation (Small et al., 2014 [\[13\]](#)). Lamellocyte induction involves the formation and sensing of reactive oxygen species triggered by wasp egg injection (Nappi et al., 1995 [\[14\]](#); Sinenko et al., 2011 [\[15\]](#); Small et al., 2014 [\[13\]](#); Louradour et al., 2017 [\[16\]](#); reviewed in: Banerjee et al., 2019 [\[1\]](#); Csordás et al., 2021 [\[8\]](#)). The molecular mechanism underlying the simultaneous crystal cell fate inhibition, however, is less well understood. Obviously, an effective and timely downregulation of Notch activity is required to fend off parasitic wasps.

The Notch pathway is highly conserved between invertebrates and vertebrates, where it regulates numerous cell fate decisions. Notch signals are transduced by CSL type proteins (abbreviation of human CBF1/RBPJ, *Drosophila* Suppressor of Hairless [Su(H)], and worm Lag1), that bind the DNA to direct transcriptional activity of Notch target genes by help of recruited cofactors (reviewed in: Giaimo et al., 2021 [\[17\]](#); Kopan and Ilagan, 2009 [\[18\]](#); Siebel and Lendahl, 2017 [\[19\]](#)). In the absence of Notch signals, however, CSL together with corepressors silences Notch target genes, thereby acting as a molecular switch (Borggrefe and Oswald, 2009 [\[20\]](#)). Hence, CSL is central to Notch pathway activity as no signal transduction can occur in its absence or in the instance of a lack of DNA-binding. Earlier, we observed Su(H) phosphorylation at Serine 269 in cultured *Drosophila* Schneider S2 cells (Nagel et al., 2017 [\[21\]](#)). Of note, Schneider S2 cells have hemocyte characteristics (Cherbas et al., 2011 [\[22\]](#); Schneider, 1972 [\[23\]](#); Terriente-Felix et al., 2013 [\[24\]](#)). As a consequence of the negative charge conferred by the phosphorylation at Serine 269, Su(H) loses its affinity to the DNA, and without its DNA-binding ability, also its function as a transcriptional regulator (Nagel et al., 2017 [\[21\]](#)). Accordingly, a phospho-mimetic *Su(H)*^{S269D} mutant behaved like a *Su(H)* loss of function

mutant in all respects. In contrast, the phospho-deficient *Su(H)*^{S269A} variant, appeared wild type at the first glance, indicating some tissue-specificity of this regulatory mechanism. In fact, the *Su(H)*^{S269A} allele displayed a gain of Notch activity particularly during embryonic and larval hematopoiesis with increased numbers of crystal cells in both hematopoietic compartments (Frankenreiter et al., 2021 [DOI](#)). Moreover, we found that the general Notch antagonist Hairless is not involved in constraining crystal cell numbers, suggesting that in the context of blood cell homeostasis Notch activity is regulated by the phosphorylation of Su(H) (Maier, 2006 [DOI](#); Frankenreiter et al., 2021 [DOI](#)). As the same set of genes affect blood cell maintenance and differentiation in homeostasis and upon immune challenge (Cho et al., 2020 [DOI](#); Tattikota et al., 2020 [DOI](#)), obviously phospho-mediated downregulation of Notch activity might also occur in response to parasitoid wasp infestation, allowing the formation of encapsulation active lamellocytes at the expense of crystal cells. Briefly, phosphorylation of Su(H) is an elegant mechanism to transiently curb Notch activity in the context of immune responses.

In this work, we followed the hypothesis that after wasp infestation, a specific kinase might be activated to phosphorylate Su(H) thereby allowing an adequate immune response. In this case, the phospho-deficient *Su(H)*^{S269A} allele should be immune-compromised, as it cannot respond to phosphorylation, i.e. remaining active even upon parasitism. Indeed, *Su(H)*^{S269A} displayed an increased sensitivity towards parasitoid wasp infestation accompanied by an increase of crystal cells at the expense of lamellocytes. Accordingly, phosphorylation of Su(H) protein was detected in infested wild type larvae but not in the *Su(H)*^{S269A} mutant. In a screen for kinases regulating this process, Pkc53E, the homologue of human PKCα, was identified as an important player. In agreement with a role in blood cell homeostasis, a *Pkc53E*^{Δ28} null mutant displayed increased crystal cell numbers. Moreover, genetic and molecular interactions between Su(H) and Pkc53E support the specificity of their relationship. Finally, *Pkc53E*^{Δ28} was impaired in its immune response to wasp infestation as well. Together, these data show that Su(H) is a target of Pkc53E during parasitic wasp infestation, inducing phosphorylation and subsequent downregulation of Su(H) activity to allow the mass production of lamellocytes required for wasp defense.

Results

Impaired immune response to parasitoid wasps in the phospho-deficient *Su(H)*^{S269A} mutant

Wasp infestation alters the course of hematopoiesis, as lamellocyte differentiation is massively increased at the expense of crystal cells. This process requires a timely attenuation of Notch activity that may be implemented by the phosphorylation of Su(H) at Serine 269 (S269). To test this model, we exposed control larvae and larvae of the phospho-deficient *Su(H)*^{S269A} variant to the parasitoid wasp *Leptopilina boulardi* (*L. boulardi*) and analysed the consequences on blood cell homeostasis around 44 hours post-infection, i.e. before lymph gland histolysis. To exclude any influence of the engineered genomic background, we used *Su(H)*^{gwt} for comparison, carrying a genomic wild type construct in place of the mutant (Praxenthaler et al., 2017 [DOI](#); Frankenreiter et al., 2021 [DOI](#)).

Earlier we noted an excess of crystal cells in the phospho-deficient *Su(H)*^{S269A} allele, which we interpret as a gain of Notch activity in consequence of the inability to be downregulated by Su(H) phosphorylation (Frankenreiter et al., 2021 [DOI](#)) (Figure 1A,B [DOI](#)). In agreement with this hypothesis, RNAi-mediated downregulation of *Notch* in hemocytes (*hml::N*-RNAi) resulted in a near complete loss of crystal cells. This *Notch* loss of function phenotype was epistatic to the *Su(H)*^{S269A} phenotype, i.e. the excess of crystal cells characterizing *Su(H)*^{S269A} was no longer observed in the combination with *N*-RNAi, demonstrating that Notch acted upstream of Su(H) as expected (Figure 1 [DOI](#) Figure supplement 1). The slightly elevated numbers in the *Su(H)*^{S269A}

background compared to the control, however, may be due to the enlarged anlagen in the embryo unaffected by *hml::N*-RNAi (Fankenreiter et al., 2021). Total hemocyte numbers were slightly, albeit not significantly increased in *Su(H)^{S269A}* compared to the *Su(H)^{gwt}* control, and correspondingly, hemocyte numbers were somewhat lowered in the *Su(H)^{S269D}* allele (Figure 1_Figure supplement 1). This is in line with earlier observations of unchanged plasmatocyte counts in *N* or *Su(H)* mutants relative to control (Duvic et al., 2002 [↗](#)). In response to wasp infestations, however, crystal cell numbers should drop to allow the formation of lamellocytes (Small et al., 2014 [↗](#); Csordás et al., 2021 [↗](#)). Indeed in the *Su(H)^{gwt}* control, both the sessile crystal cells as well as those within the larval lymph glands were significantly lessened in response to wasp infestation (**Figure 1A,B** [↗](#)). In contrast, the higher crystal cell numbers in the *Su(H)^{S269A}* mutant larvae dropped to control-level, demonstrating the impairment of the mutant to detect this immune challenge or to respond to it (**Figure 1A,B** [↗](#)). Total hemocyte numbers, however, were similar between the genotypes independent of wasp infestation (Figure 1_Figure supplement 1).

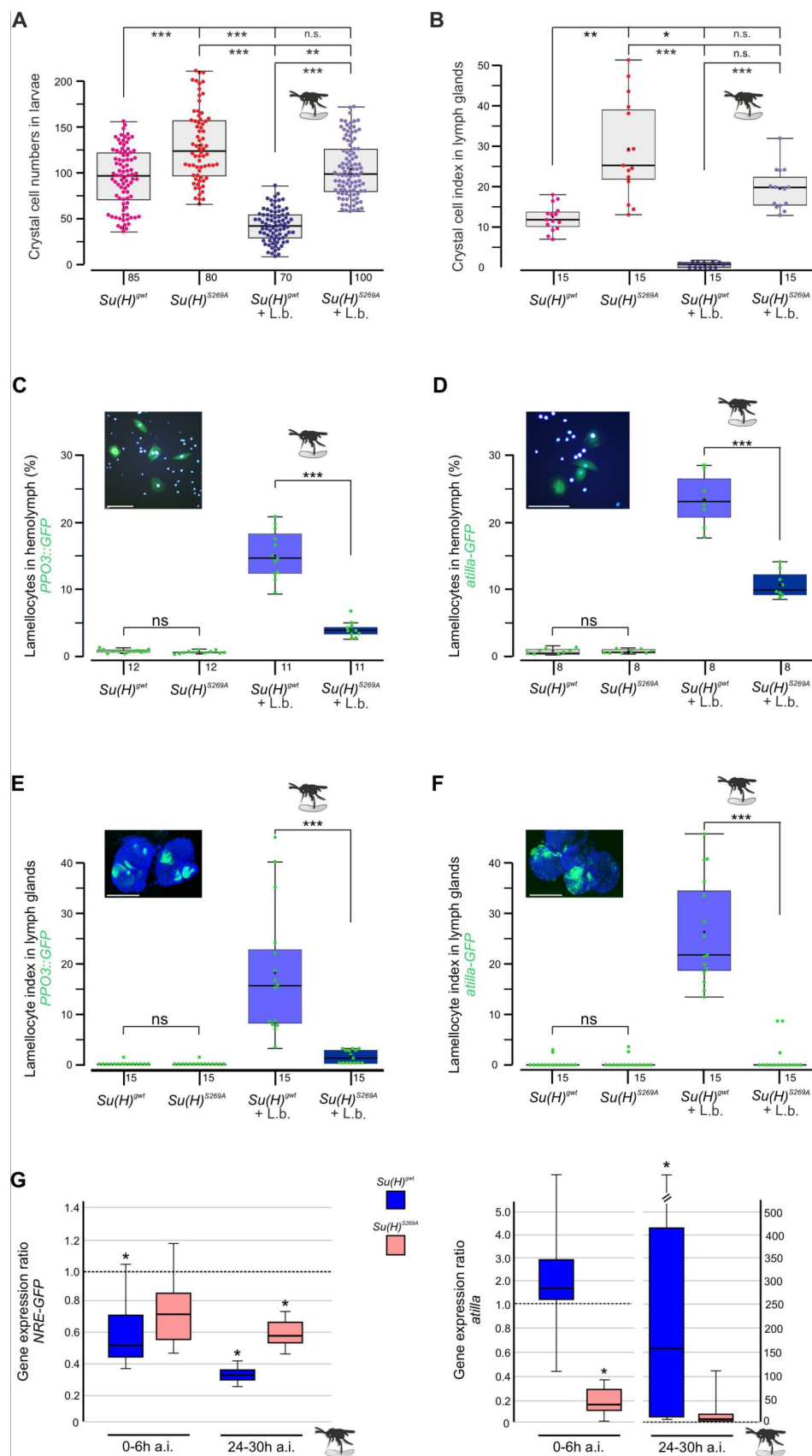


Figure 1

***Su(H)*^{S269A} mutants are compromised in their response to parasitoid wasp infestation**

(A) Quantification of melanized crystal cells from the last two segments of *Su(H)*^{gwt} and *Su(H)*^{S269A} larvae with and without wasp infestation as indicated. In the control, wasp parasitism causes crystal cell numbers to drop to a level of about 50%, whereas in *Su(H)*^{S269A} mutants the number settles at the un-infested *Su(H)*^{gwt} level. Each dot represents one analysed larva (n=70-100 as indicated). (B) Crystal cell index in larval lymph glands is given as ratio of Hnt-positive crystal cells per 1° lobe relative to the size of the lobe. Each point represents one analysed lobus (n=15). (A,B) Statistical analyses Kruskal-Wallis test, followed by Dunn's test with *** p<0.001, ** p<0.01, * p<0.05, ns (not significant p≥0.05).

(C-F) Quantification of larval lamellocytes in the circulating hemolymph (C,D) or in lymph glands (E,F) before and after wasp infestation in *Su(H)*^{gwt} versus *Su(H)*^{S269A}. Lamellocytes were marked with either *PPO3-Gal4::UAS-GFP* (C,E) or *atilla-GFP* (D,F) as indicated. (C,D) The fraction of GFP-labelled lamellocytes of the total number of DAPI-labelled blood cells isolated from hemolymph is given; each dot represents ten pooled larvae (n=8-10 as shown). Representative image of labelled control hemolymph is shown above (DAPI-labelled nuclei in light blue, GFP in green). Scale bars 50 µm. (E,F) Lamellocyte index is given as number of GFP-labelled lamellocytes per area in the 1° lobe of the lymph gland. Each dot represents the lamellocyte index of one lobus (n=15). Representative *Su(H)*^{gwt} lymph glands after wasp infestation are shown, co-stained for nuclear PzG (in blue). Scale bars 100 µm. Statistical analyses with unpaired Student's t-test; only significant differences are indicated (*** p<0.001). (A-F) Representative images for each genotype and condition are shown in supplemental Figure 2.

(G) qRT-PCR analyses measuring expression of NRE-GFP (left panel) and *atilla* (right panel). Transcript levels were quantified from hemolymph isolated from infested larvae at 0-6 h or 24-30 h post-infestation as indicated, relative to the untreated *Su(H)*^{gwt} control. *Tbp* and *cyp33* served as reference genes. Shown data were gained from four biological and two technical replicates each. Left panel: Immediately after wasp infection, NRE-GFP expression dropped significantly in the *Su(H)*^{gwt} control, and even further to about 30% 24-30 h post-infection, whereas it remained at 60%-70% in the infested *Su(H)*^{S269A} mutants. Right panel: *atilla* transcripts remained stable at first in the *Su(H)*^{gwt} control, to rise dramatically 24-30 h post-infection, in contrast to *Su(H)*^{S269A}. Mini-max depicts 95% confidence, mean corresponds to expression ratio. Exact p-values are given in the raw data table. Significance was tested using PFRR from REST (*p<0.05).

Increased abundance of lamellocytes upon wasp infestation was monitored *in vivo* in larval hemolymph and lymph glands, using either the L1-*atilla*-GFP reporter (Honti et al., 2009) or *PPO3-Gal4::UAS-GFP* (Dudzic et al., 2015), respectively. In the harvested larval hemolymph of the infested *Su(H)*^{gwt} control, the hemocytes contained about 15% *PPO3*- and 23% *atilla*-labelled lamellocytes (Figure 1C,D). These numbers were significantly lower in the wasp infested *Su(H)*^{S269A} larvae (Figure C,D). Consistently, both lamellocyte reporters were robustly induced in the lymph glands of the infested control, but rarely in glands of the *Su(H)*^{S269A} mutant (Figure 1E,F and Figure1_supplement 2). Obviously, the *Su(H)*^{S269A} mutant barely responds to the immune challenge raised by the parasitic wasp infestation.

In order to monitor the altered immune responses at the molecular level, we quantified Notch target gene expression in the hemolymph upon wasp infestation over time. We observed a decline in the expression of the Notch reporter NRE-GFP immediately after wasp infection in the control *Su(H)*^{gwt} to about half the value of the uninfected larvae, dropping even further to about 30% 24-30 hours post infection (Figure 1G). *Su(H)*^{S269A} mutant larvae, however, retained a much stronger expression level at around 60-70% of the uninfected control even at the late time point. These data reveal the downregulation of Notch activity in response to wasp infestation prior or parallel to lamellocyte formation, in agreement with our model, whereby the infestation-induced

phosphorylation of Su(H) impairs transmission of Notch signalling activity. Accordingly, the lamellocyte specific marker *atilla* bounced up nearly two magnitudes in the wasp infected *Su(H)^{gwt}* control at the late time point, but not in *Su(H)^{S269A}* compared to the uninfected control (**Figure 1G**) (Cattenoz et al., 2020).

Overall, these data support the model that S269 in Su(H) is a molecular target for a kinase, phosphorylated upon immune challenge to allow lamellocyte formation at the expense of crystal cells.

The phospho-deficient *Su(H)^{S269A}* allele is impaired in combating wasp infestation

Parasitoid wasp infestation constitutes an extreme immune challenge for the *Drosophila* larva: if not combatted by the immune system, a wasp egg, which is deposited in the larval body cavity, will develop into an adult wasp, thereby killing the larval host during the pupal stage. Indeed, depending on the wasp species used, we measured a high mortality rate with less than 5% up to about 14% of surviving flies, whereas nearly all pupae hatched to adults without wasp challenge (**Figure 2A**). According to our working hypothesis, the phospho-deficient *Su(H)^{S269A}* variant should not be able to properly respond to the immune challenge. To test this directly, we measured the survival of *Su(H)^{S269A}* animals upon wasp infestation compared to the wild type control.

The two closely related wasp species *L. boucardi* and *Leptopilina heterotoma* (*L. heterotoma*) very efficiently parasitized both the control *Su(H)^{gwt}* and the *Su(H)^{S269A}* variant, though the latter appeared slightly more sensitive (**Figure 2A**). The difference in mortality became more apparent with the wasp species *Asobara japonica* (*A. japonica*), allowing 14% of the *Su(H)^{gwt}* control flies to escape parasitism, whereas only 3.4% of the infested *Su(H)^{S269A}* pupae emerged as flies. Thus, the *Su(H)^{S269A}* mutants are less robust in resisting parasitoid wasp infestation consistent with an impaired immune response.

Serine 269 of Su(H) is phosphorylated upon wasp infestation

Next, we wanted to directly monitor Su(H) phosphorylation at S269 in response to wasp infestation. To this end, polyclonal antibodies directed against a phosphorylated peptide containing the sequence motif NRLRpSQTVSTRYLHVE were generated (α -pS269). Specificity was first tested by Western blot analysis using bacterially expressed GST fusion proteins containing the entire beta-trefoil domain (BTD) of Su(H), as well as with phospho-mimetic (S269D) and phospho-mutant (S269A) versions. All three variants were detected by the antisera. The S269D version, however, was strongly preferred, indicating that this antibody does preferably recognize phospho-S269 Su(H) protein (**Figure 2** - Figure supplement 1). Encouraged by this result, we used this antiserum on protein extracts derived from larvae infested and not infested by *L. boucardi*. For this experiment we used genome engineered fly strains expressing mCherry-tagged Su(H) proteins, *Su(H)^{S269A-mCh}* and *Su(H)^{gwt-mCh}* for control (Praxenthaler et al., 2017). Su(H) proteins were trapped using RFP-nanobody coupled agarose beads, and the precipitates were then probed in Western blots with antibodies directed against mCherry and pS269. Indeed, α -pS269 antibodies recognized *Su(H)^{gwt-mCh}* protein in wasp infested larvae, indicating respective phosphorylation of Su(H) protein. No such signals were seen in precipitates from the non-infested larvae nor from any of the *Su(H)^{S269A-mCh}* mutant larvae (**Figure 2B**). These data strongly support the notion of parasitism-induced phosphorylation of Su(H) at S269. Apparently, parasitoid wasp infestation starts a cascade of events resulting in the inhibitory S269 phosphorylation of Su(H) protein to allow lamellocyte formation. Hence, the question arose on the kinase/s involved in this process.

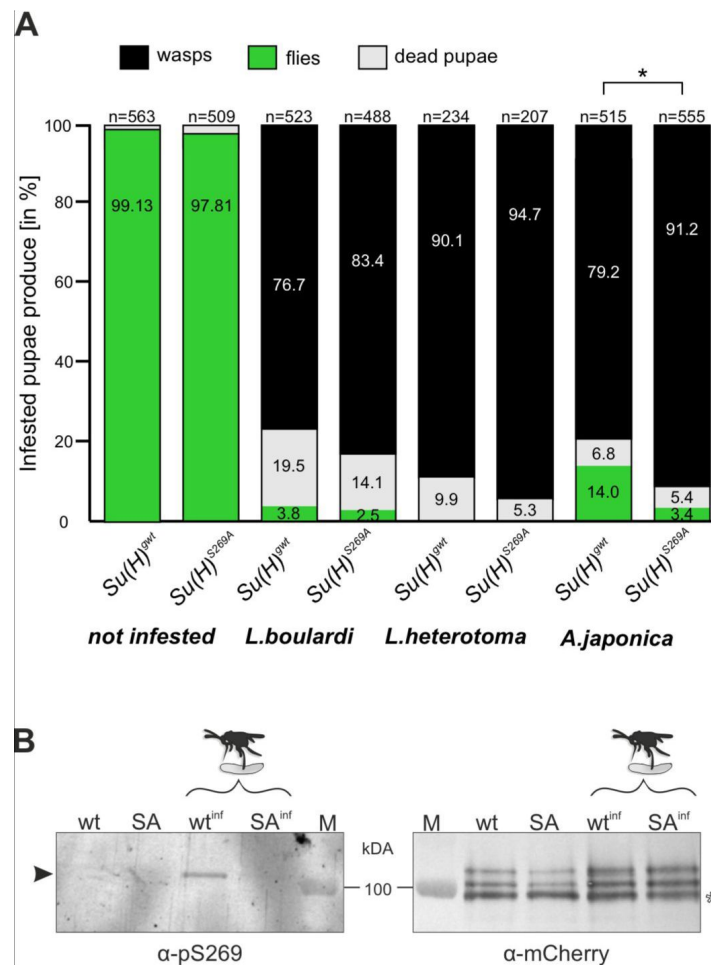


Figure 2

Resistance to wasp infestation and phosphorylation at S269

(A) Resistance of *Su(H)*^{gwt} and *Su(H)*^{S269A} to the infestation with parasitic wasp strains *L. boulandi* and *L. heterotoma* (both family Figitidae) and *A. japonica* (family Braconidae), as indicated. Numbers of eclosed flies versus wasps as well as of dead pupae are presented in relation to the total of infested pupae. Left two columns are from non-infested controls. At least three independent experiments were performed, n=number of infested pupae. Statistically significant difference determined by Student's t-test is indicated with *p<0.05.

(B) *Su(H)* is phosphorylated at Serine 269 upon parasitoid wasp infestation. Protein extracts from *Su(H)*^{gwt-mCh} (wt) and *Su(H)*^{S269A-mCh} (SA) larvae, respectively, infested (wt^{inf}, SA^{inf}) with *L. boulandi* or uninfested, were isolated by RFP-Trap precipitation and probed in Western blots. The anti-pS269 antiserum specifically detects wild type *Su(H)* protein only in wasp infested larvae (arrowhead), but not the *Su(H)*^{S269A} isoform. The blot on the right served as loading control, probed with anti-mCherry antibodies, revealing the typical *Su(H)* protein pattern in all lanes; the lowest band presumably stems from degradation (open asterisk). M, prestained protein ladder, protein size is given in kDa.

Screening of kinase candidates mediating phosphorylation of Su(H) at Serine 269

To identify Ser/Thr kinases involved in the phosphorylation of Su(H) at S269, we commenced with a combination of *in silico* and biochemical approaches aiming to generate a list of candidate kinases which can be further analysed by genetic means (Figure 3). First, by using GPS 3.0 software that encompasses a substantial database of kinases and their preferred recognition motives (Xue et al., 2011), 36 potential human kinases were predicted to recognize S269 as substrate, represented by 30 kinases in *Drosophila* (Figure 3 –Table 1 supplement). In addition, the BTD domain of Su(H) was bacterially expressed as a GST fusion protein and subjected to phospho-assays using 245 different human Ser/Thr kinases. Our reasoning for using the entire BTD domain was to ensure a normal folding of the domain (Kovall and Blacklow, 2010), and to reduce the number of potential phospho-sites at the same time present in full length Su(H). With this approach, we ended up with 62 human Ser/Thr kinases (25% of the tested kinases) that use the BTD of Su(H) as an *in vitro* substrate, corresponding to 40 different kinases in *Drosophila* (Figure 3 –Table 2 supplement). Both sets of data, biochemical and bioinformatics, were used to generate a list of 44 candidates to be analysed by genetic means. The candidates were further screened for an imbalanced hematopoiesis. We reasoned that mutants affecting a relevant kinase gene involved in the phosphorylation of Su(H) should display increased crystal cell numbers similar to what was observed in the *Su(H)*^{S269A} mutant (Frankenreiter et al., 2021). Larvae of 13 different kinase mutants, and the progeny of 44 UAS-RNAi and/or UAS-kinase dead transgenes crossed with *hml-Gal4*, a blood cell-specific Gal4 driver line, were tested (Figure 3 –Table 3 supplement). To this end, the larvae were subjected to heating for a visualization and quantification of sessile crystal cells (Rizki, 1957; Lanot et al., 2001). About a third of the kinase mutants were similar to the control, whereas mutations in four kinases impeded crystal cell development or prevented it altogether. Unexpectedly, the majority of the tested kinase mutants exhibited elevated crystal cell numbers, however to a different degree (Figure 3_Figure supplement 1, Table 4 supplement) (Deichsel et al., 2024). Nineteen kinase mutants matched closely the *Su(H)*^{S269A} phenotype, making those the most promising candidates to being involved in the phosphorylation of Su(H) at S²⁶⁹. Six of those were within the cluster of ten candidates singled out by the *in silico* and the *in vitro* screens (Figure 3 A,B). Using commercially available, activated human kinases, we were able to test five candidates, AKT1, CAMK2D, GSK3B, S6 and PKCα *in vitro* by MS/MS analysis on the Su(H) peptide ALFNRLRS⁸QTVSTRY, where Serine 8 (S8) corresponds to S269 in Su(H). Only PKCα unambiguously phosphorylated the given peptide at Serine 8. Whereas GSK3B did not phosphorylate the peptide at all, AKT1, CAMK2D and S6 piloted Threonine 10, corresponding to Threonine 271 in Su(H) (Figure 3D, Figure 3_Figure supplement 2). Together, these data support the idea, that PKCα corresponding to Pkc53E in *Drosophila*, is part of the kinase network mediating the phosphorylation of Su(H) at S269.

Role of Pkc53E in the phosphorylation of Su(H)^{S269}

Confirming the MS/MS data, human PKCα was able to phosphorylate the respective Su(H) peptide (S^{WT}) similar to its defined pseudo-substrate PS (Kochs et al., 1993), whereas the S8A mutant peptide (S^{SA}) was accepted only half as well in an ADP-GloTM assay, indicating that S269 is a preferred substrate (Figure 4A). Bacterially expressed and purified *Drosophila* Pkc53E, however, did neither accept the PS nor the Su(H) peptides (Figure 4B). Pkc53E activity, however, was stimulated by the agonistic phorbol ester PMA (phorbol 12-myristate 13-acetate) (Blumberg et al., 1983; Nakashima, 2002) to phosphorylate the PS and Su(H) peptide S^{WT} but not the S8A mutant peptide S^{SA} (Figure 4C). To generate an activated form of Pkc53E, we exchanged four codons by *in vitro* mutagenesis, three (T508D, T650D and S669D) mimicking phosphorylation in the kinase and C-terminal domains, respectively, and one in the pseudo-substrate domain (A34E) (Figure 4_Figure supplement 1) (Gould and Newton, 2008). The resultant Pkc53E^{EDDD} kinase accepted the PS, and the Su(H) peptide S^{WT} even better, but not the S8A mutant peptide S^{SA},

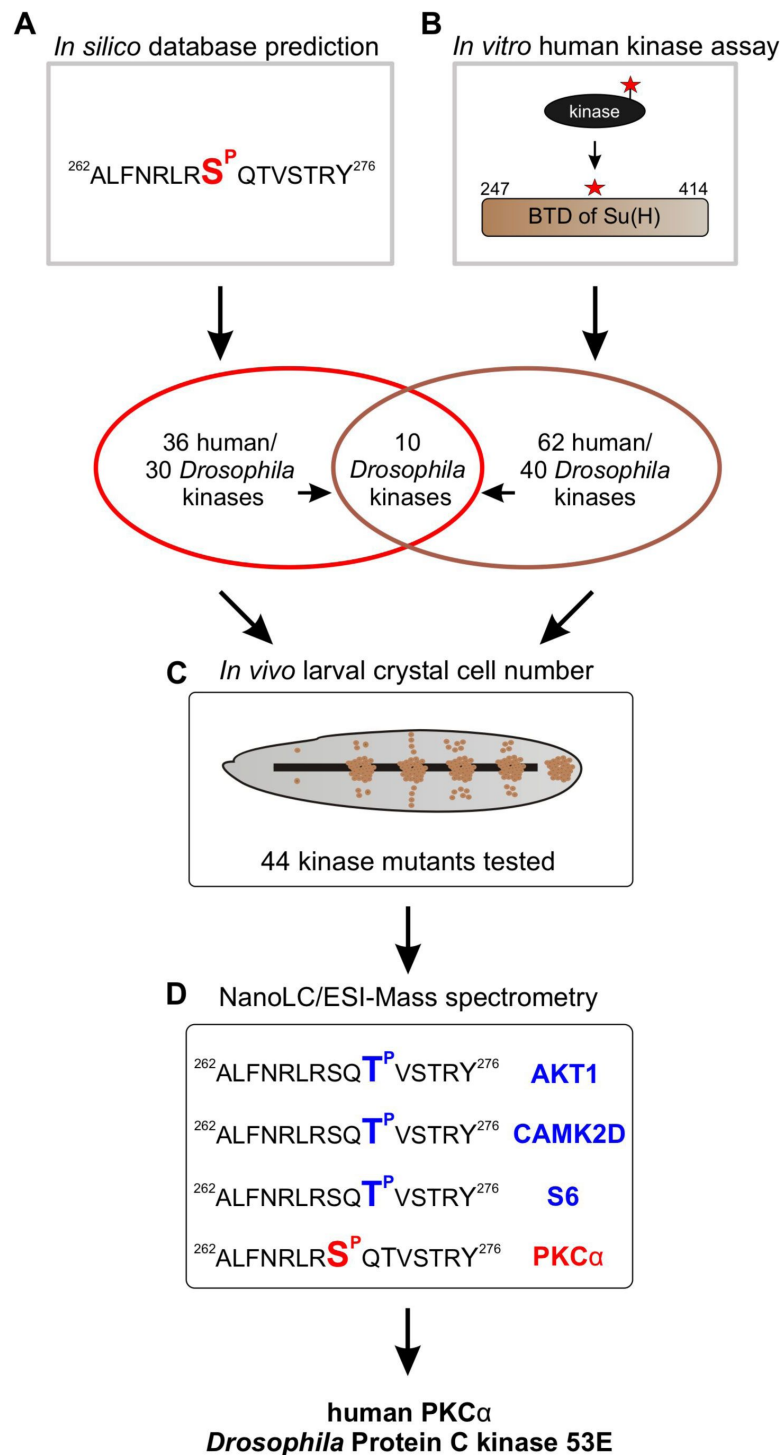


Figure 3

Pipeline of screening procedures for kinase candidates triggering phosphorylation at Su(H)^{S269}

(A) *In silico* screening of database(s) predicting kinase recognition motif in Su(H)^{S269}; see Table 1 supplement. (B) *In vitro* assay screening 245 human Ser/Thr kinases for their ability to phosphorylate the BTD domain of Su(H); see Table 2 supplement. (C) *In vivo* screen of 44 different *Drosophila* kinase mutants for crystal cell occurrence in third instar larvae; see Table 4 supplement. (D) NanoLC/ESI mass spectrometry with active human kinases monitoring their ability to phosphorylate the given Su(H) peptide. PKC α phosphorylates S269, whereas AKT1, CAMK2D and S6 kinase prefer T271. Spectra are shown in Figure 3 supplement 2.

demonstrating the specificity of the phosphorylation (**Figure 4D** [↗](#)). As predicted for a fully activated kinase, PMA was unable to boost Pkc53E^{EDDD} protein activity any further (**Figure 4E** [↗](#)). The bacterially expressed Pkc53E^{EDDD} kinase, however, was a magnitude less active compared to commercial PKC α , perhaps reflecting poor quality of the bacterially expressed protein, or indeed an intrinsic biochemical property of the *Drosophila* enzyme.

The PKC-agonist PMA influences Su(H) activity and blood cell homeostasis

Our data so far indicated that Su(H) is a phospho-target of Pkc53E which reduces its activity by affecting its DNA-binding. In this case, we might expect an influence of the general PKC-activator PMA on both, Su(H) activity as well as Notch-mediated crystal cell formation. We tested the former in a *RBPJ^{ko}* HeLa cell system (Wolf et al., 2019 [↗](#)), measuring Notch-reporter gene activation by Su(H)-VP16. To this end, a Su(H)-VP16 gene fusion was cloned under HSV-TK promoter control, which is unresponsive to PMA in HeLa cells (Shifera and Hardin, 2009 [↗](#)). Su(H)-VP16 protein is independent of Notch activity itself, allowing to directly monitor the influence of PMA on Su(H) activity. Indeed, Su(H)-VP16 ability to activate reporter gene transcription was reduced by more than half in the presence of PMA (**Figure 5A** [↗](#)). This is in agreement with a PMA-mediated activation of endogenous PKC in the transfected HeLa cells, resulting in Su(H)-VP16 phosphorylation, loss of DNA-binding activity and reduced transcriptional activation. Accordingly, repression could be reverted by the addition of kinase-inhibitor Staurosporine (Karaman et al., 2008 [↗](#)). In fact, Staurosporine alone already increased Su(H)-VP16 transcriptional activity, suggesting that inhibitory phosphorylation of Su(H) occurs in HeLa cells (**Figure 5A** [↗](#)). Expression levels of Su(H)-VP16, however, remained unchanged by the treatments (Figure 5_Figure supplement 1).

Next, we assayed the effect of PMA on larval crystal cell formation. If, as expected, PMA increased Pkc53E activity, Su(H) should be inactivated by phosphorylation with decreased crystal cell numbers as a consequence. We fed PMA to *Drosophila* larvae and assayed the numbers of sessile crystal cells. In agreement with our expectations, crystal cell numbers dropped very strongly, suggesting efficient phosphorylation and inactivation of Su(H) protein by PKCs (**Figure 5B** [↗](#)). As predicted by the above experiments, this effect was alleviated by Staurosporine. Owing to the global inhibition of kinases by Staurosporine, however, a rise in crystal cells was expected, because many kinases restrict their numbers (see Figure 3_Figure supplement 1) (Deichsel et al., 2024 [↗](#)). In contrast, *Su(H)^{S269A}* mutant larvae displayed an increased number of crystal cells which can be attributed to the fact that here, Su(H) can no longer be phosphorylated and hence, is overactive in this context. Feeding PMA to *Su(H)^{S269A}* larvae caused only a minor drop of excessive crystal cell numbers, which could be due to other kinases acting on *Su(H)^{S269}* or due to other Pkc53E substrates involved in crystal cell development. Again, kinase inhibition by Staurosporine largely reversed this effect (**Figure 5B** [↗](#)). In conclusion these data support the notion that Su(H) activity is regulated *in vitro* and *in vivo* by PKC activity in the context of blood cell homeostasis.

Pkc53E is required for normal blood cell homeostasis in *Drosophila* larvae

Su(H)^{S269A} mutant larvae develop an excess of crystal cells, both in the hemolymph as well as in the lymph glands, due to a failure to downregulate respective Notch activity in the particular progenitor cells via the phosphorylation of Su(H) protein (Frankenreiter et al., 2021 [↗](#)) (see **Figure 1A,B** [↗](#)). Assuming Pkc53E has a major role in the phosphorylation of Su(H), we should expect a similar phenotype in a *Pkc53E* mutant due to the inability to phosphorylate any substrate. In order to test this assumption directly, we assessed the number of crystal cells in larval lymph glands as well as in sessile crystal cells in several loss of function backgrounds of *Pkc53E*. To this end, we used the *Pkc53E^{A28}* allele, which by RT-PCR is a null mutant (**Figure 6** [↗](#) – Figure supplement 1). In

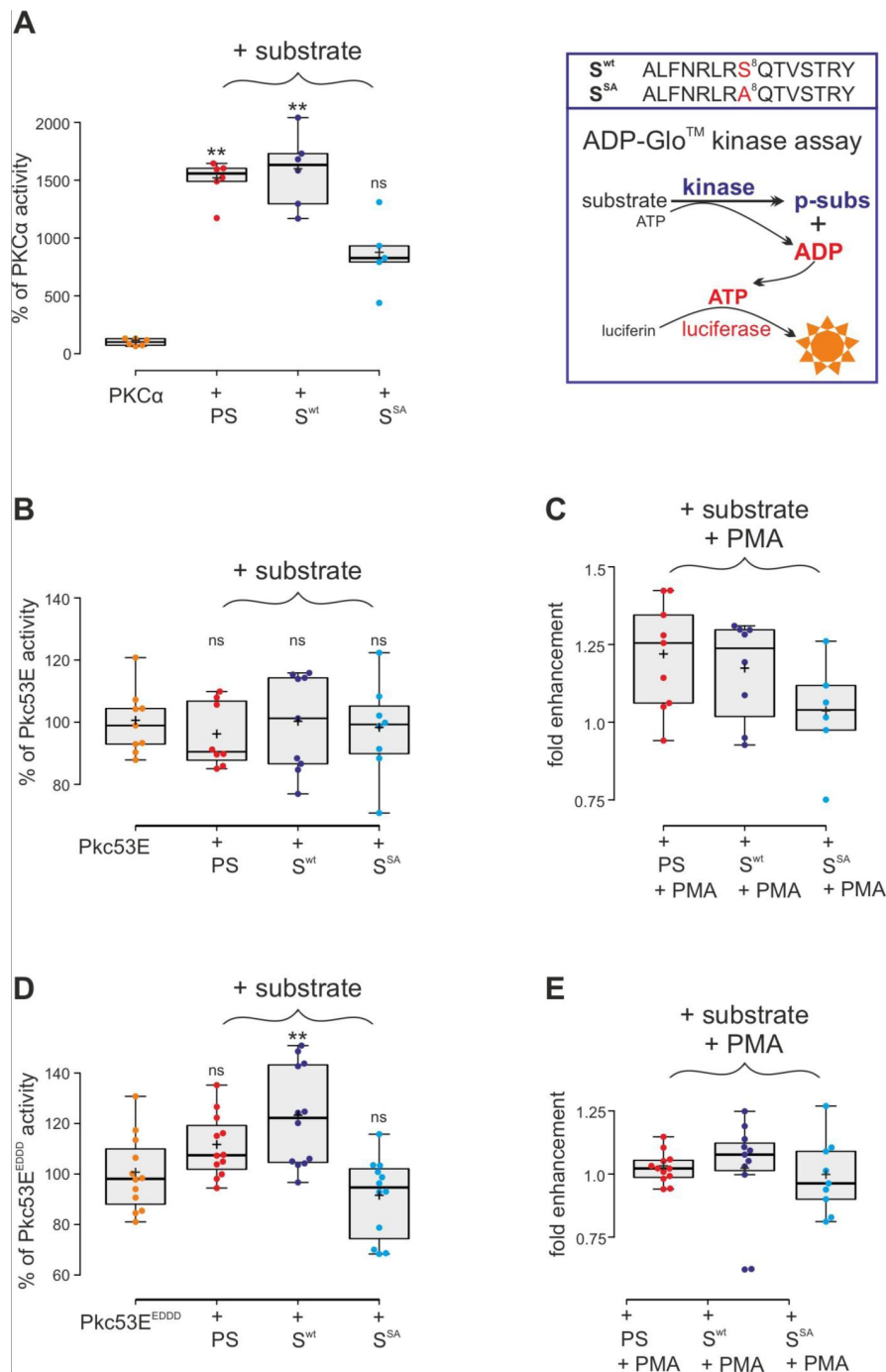


Figure 4

Kinase assays using activated PKCα and Drosophila Pkc53E variants

(A) Right, schema of ADP-Glo™ assay to quantify kinase activity. The wild type (S^{wt}) and mutant (S^{SA}) Su(H) peptides 262-276 offered as specific kinase substrates are indicated above. Left, the commercially available, active human PKCα very efficiently phosphorylates the pseudo-substrate PS and the Su(H) S^{wt} peptide, but less efficiently the S^{SA} mutant peptide. Activity is given as percentage of the auto-active kinase without substrate. Each dot represents one experiment. **(B)** Bacterially expressed Pkc53E has no activity on any of the offered substrates PS, S^{wt} or S^{SA}. **(C)** PMA raised Pkc53E activity to nearly 125% for PS and S^{wt} but not for S^{SA}. **(D)** Activated Pkc53^{EDDD} phosphorylates PS and S^{wt} but not of S^{SA}. **(E)** Addition of PMA does not change Pkc53^{EDDD} activity. Statistical analyses were performed with Kruskal-Wallis test followed by Dunn's test (A,C,E) or ANOVA followed by Tukey's approach in (B,D) with ** p<0.01, ns (not significant p≥0.05).

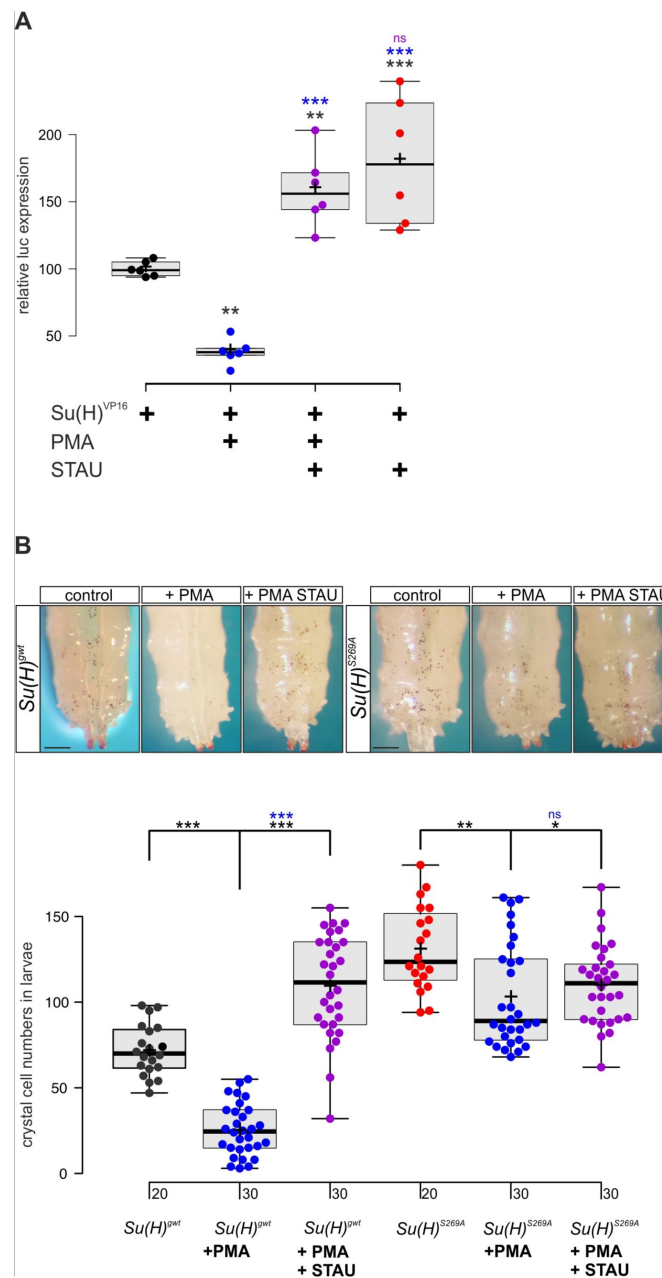


Figure 5

PMA inhibits Su(H) transcriptional activity in vitro and crystal cell formation in vivo

(A) Expression of NRE-luciferase reporter gene in *RBP1^{ko}* HeLa cells, transfected with 2xMyc-Su(H)-VP16 [Su(H)^{VP16}]. Luciferase activity is given relative to the reporter construct normalized to Su(H)-VP16 set to 100%. Addition of PMA causes reduction of Su(H)-VP16 dependent transcriptional activity to about 40%, which is reversed by the kinase inhibitor Staurosporine (STAU). STAU itself results in increased Su(H)-VP16 activity. Each dot represents one experiment (n=6). Statistical analysis was performed with ANOVA followed by Dunnet's multiple comparison test with *** p<0.001, ** p<0.01 relative to Su(H)-VP16 alone (black asterisks) or to Su(H)-VP16 plus PMA (blue asterisks).

(B) Number of melanized larval crystal cells determined in the last two segments of larvae fed with fly food plus 10% DMSO (control), or with fly food supplemented with 1mM PMA, or 1 mM PMA plus 0.2 mM STAU (n=20 or 30 as indicated). Note strong drop of crystal cell numbers in the *Su(H)^{wt}* control fed with PMA, and a reversal by STAU addition even above control levels. In contrast, PMA has a small effect on crystal cell number in the *Su(H)^{S269A}* mutant, which is reversed by STAU. Representative animals are shown above; scale bar 250 μ m. Statistical analysis was performed by ANOVA followed by Tukey's multiple comparison test (*** p<0.001), significant differences are colour coded.

addition, we used two different RNAi-lines and one sgRNA line under UAS-control to knockdown *Pkc53E* activity specifically in progenitor cells within the developing lymph gland using *lz*-Gal4 and in the hemolymph using *hml*-Gal4, respectively (Lebestky et al., 2000). As the *hml*-Gal4 driver is active in plasmatocytes and pre-crystal cells (Mukherjee et al., 2011; Tattikota et al., 2020), it should affect *Pkc53E* activity prior to crystal cell commitment in the hemolymph. However, within the lymph gland, *hml* appears specific to the plasmatocyte lineage and not present in crystal cell precursors. Instead, we choose *lz*-Gal4 for the *Pkc53E* knockdown, as *lz* is expressed in differentiating crystal cells of the lymph gland (Lebestky et al., 2000; Blanco-Obregon et al., 2020).

In any context tested, the number of crystal cells was strongly increased matching those of the *Su(H)^{S269A}* mutant (Figure 6, Figure 6_supplement 2). The similar phenotypes imply that *Pkc53E* acts through the phosphorylation of Su(H) specifically within hemocytes to restrict crystal cell differentiation. However, as outlined above, the majority of kinase mutants displayed increased crystal cell numbers, raising the possibility of a fortuitous accordance. If the increase of crystal cell numbers in *Pkc53E* mutants is independent of Su(H) phosphorylation, we should expect an additive effect if we combine the two mutants. The double mutants *Su(H)^{S269A} Pkc53E^{Δ28}* were generated by genetic recombination; they displayed the same range of excessive crystal cell numbers as the single mutants (Figure 7A,B, Figure 7_supplement 1). Moreover, the strongly reduced number of crystal cells observed in the *Su(H)^{S269D}* mutant was not increased by *Pkc53E^{Δ28}* (Figure 7A,B), indicating that *Pkc53E* indeed acts upstream of Su(H), or directly on Su(H). If the latter is the case, we may expect the two proteins to form complexes *in vivo*. Indeed, we could co-precipitate Su(H)-*Pkc53E* protein complexes, both from *Drosophila* heads containing hemocytes (Sanchez Bosch et al., 2019), as well as from the larval hemolymph (Figure 7C,D). Specific co-precipitation was eased by using fly strains expressing m-Cherry tagged Su(H) (Praxenthaler et al., 2017) and HA-tagged *Pkc53E*, respectively. Together, these data demonstrate that *Pkc53E* has an important role in blood cell homeostasis that can be largely explained by its activity to phosphorylate Su(H), thereby regulating the activity of Notch target genes during hemocyte and lymph gland development.

Pkc53E mutants are immune compromised

According to our hypothesis, *Pkc53E* phosphorylates Su(H) in response to an immune challenge by parasitic wasp infestation. In fact, a *Pkc53E*-eGFP fusion protein expressed from the *Pkc53E* locus via protein trap (Lee et al., 2018) was observed in the cytoplasm of all blood cell types independent of wasp infestation (Figure 8). Moreover, *Pkc53E* mRNA was expressed of *Pkc53E* function should affect the ability of *Drosophila* larvae to fight wasp infestations similarly to the *Su(H)^{S269A}* mutant. This was indeed the case. Firstly, the *Pkc53E^{Δ28}* null mutant was likewise impaired in fighting an infestation with the wasp *A. japonica* as was the *Su(H)^{S269A}* mutant or the double mutant *Su(H)^{S269A} Pkc53E^{Δ28}* with only about 4% surviving flies versus 14% in the control (Figure 9A). In contrast to *Su(H)^{S269A}*, however, the *Pkc53E^{Δ28}* mutant larvae contained significantly lower hemocyte numbers independent of infestation, perhaps partly explaining the poor immune response (Figure 9_supplement S1) (McGonigle et al., 2022). Without infestation, however, *Pkc53E^{Δ28}* mutant larvae developed normally to adulthood (Figure 9_supplement S1). Moreover, when *Pkc53E^{Δ28}* was infested with the parasitic wasp *L. boulardi*, lamellocyte numbers in the hemolymph did not reach wild type levels, and they were almost absent from the larval lymph glands (Figure 9B,C). Apparently, the *Pkc53E^{Δ28}* null mutant is impaired in recognizing parasitic wasp infestation or is hampered responding to it, for example by the phosphorylation of Su(H), demonstrating the involvement of this kinase in the immune response of *Drosophila* to parasitoid wasp infestation.

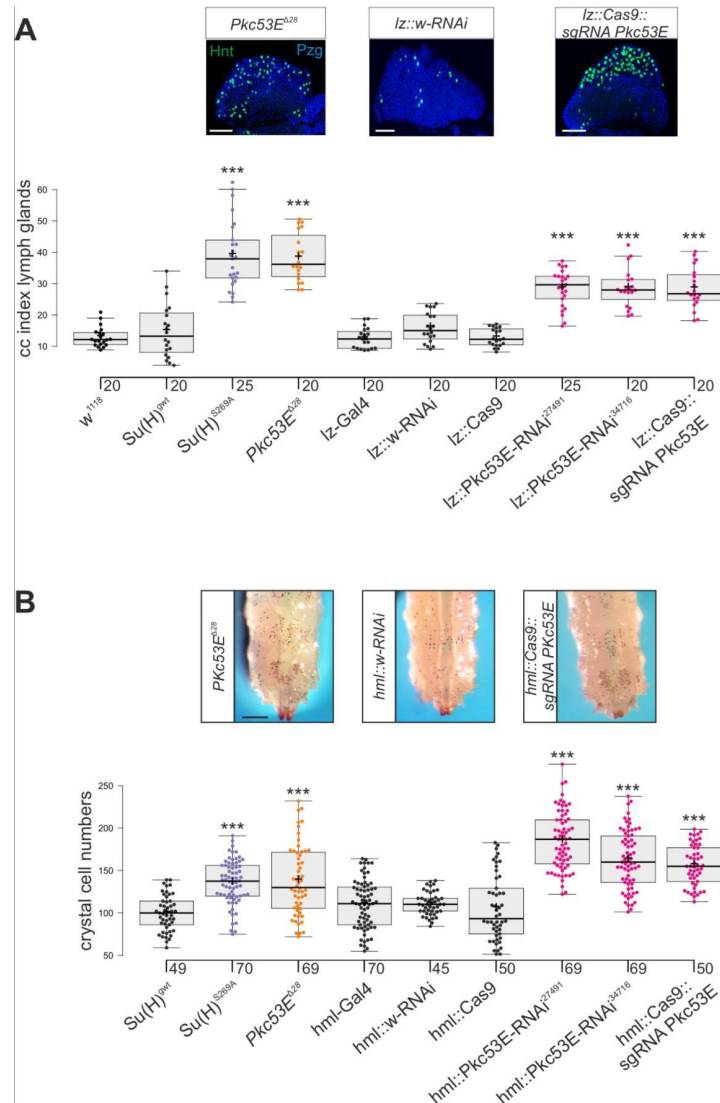


Figure 6

Loss of *Pkc53E* causes a gain of crystal cell number

Depletion of *Pkc53E* activity in the *Pkc53E^{Δ28}* mutant or after knockdown by *Pkc53E*-RNAi or *sgPkc53E* with the help of the Gal4-UAS system using *Iz-Gal4* (A) or *hml-Gal4* (B). Controls as indicated. **(A)** Crystal cell index in lymph glands; each dot represents the value of an analysed lobus (n> 20 as indicated). Examples of *Pkc53E^{Δ28}*, *Iz::Pkc53-RNAi* and *Iz::Cas9 sgPkc53E* are shown above. Crystal cells are labelled with Hnt (green), the lobe is stained with α-Pzg (blue). Scale bar 50 μm. Representative images of lymph glands for each genotype are shown in supplemental Figure S2A. Statistical analysis by ANOVA followed by Tukey's multiple comparison test relative to controls with *** p<0.001.

(B) Melanized crystal cells enumerated from the last two segments of larvae with the given genotype (n=45-70 as indicated). Examples of respective *Pkc53E^{Δ28}*, *Iz::Pkc53-RNAi* and for each genotype are shown in supplemental Figure S2B. Statistical analysis by Kruskal-Wallis test, followed by Dunn's test relative to controls with *** p<0.001. Note that there were no significant differences between any of the controls shown in black.

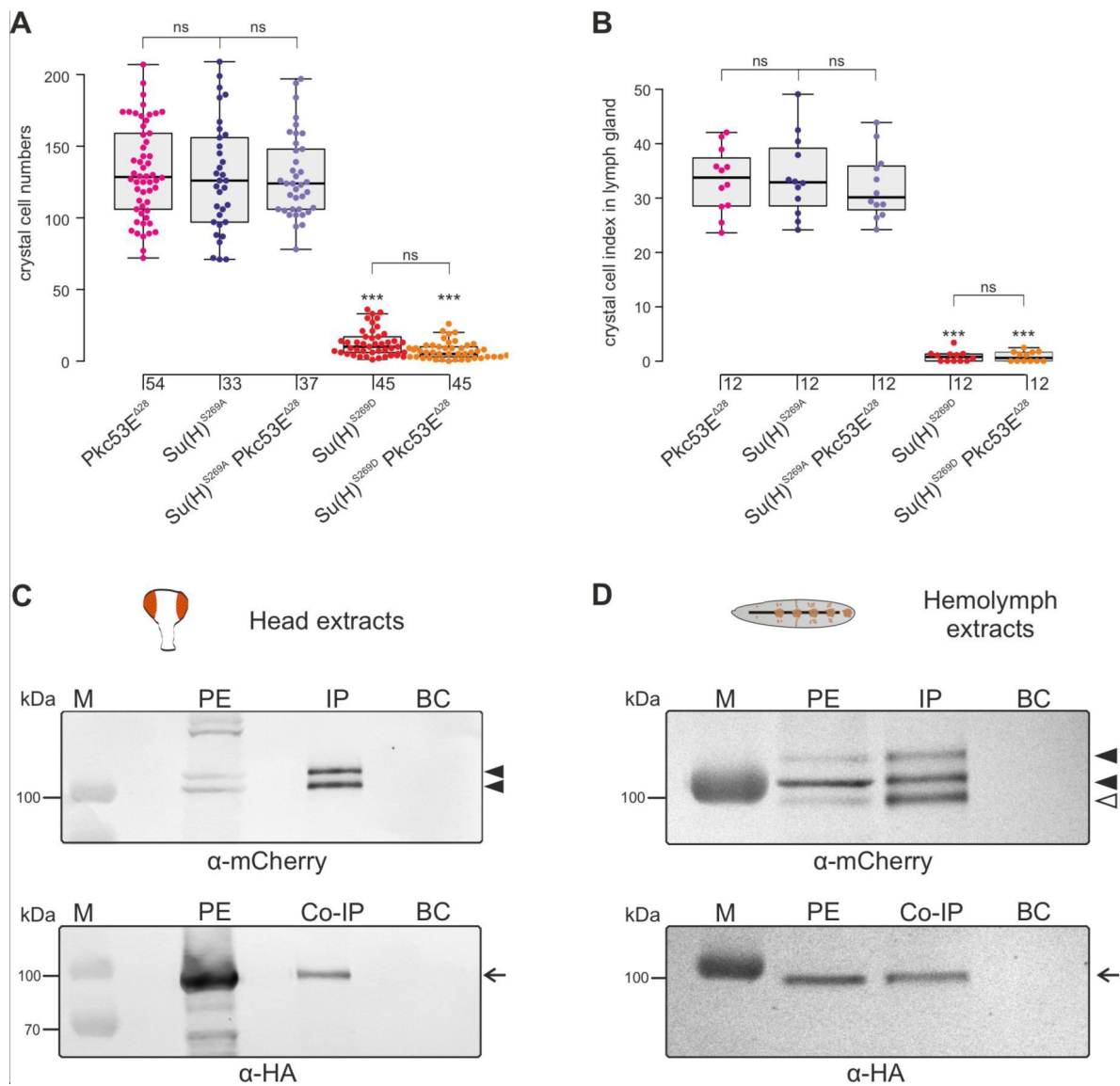


Figure 7

Pkc53E interacts with Su(H) at a genetic and a physiological level

(A) Larval crystal cell numbers and **(B)** crystal cell indices in lymph glands were determined in the given genotypes. Each dot represents one analysed larva (n, as indicated) (A) or lymph gland lobus (n=12) (B). Statistical analysis by Kruskal-Wallis test, followed by Dunn's test relative to controls with *** p<0.001; p≥0.05 ns (not significant). Representative images of sessile crystal cells and of lymph glands for each genotype are shown in Figure supplement S1.

(C,D) Co-immuno-precipitation of Pkc53E^{HA} with Su(H)^{gwt-mCh} protein. RFP-Trap IP was performed with protein extracts from 400 heads (C) or 25 third instar larvae (D), respectively. UAS-Pkc53E-HA expression was induced with *Gmr*-Gal4 in the head or with *hml*-Gal4 in the hemolymph. Endogenous mCherry-tagged Su(H) was trapped and detected with anti-mCherry antibodies (black arrowheads). The lowest band from the hemolymph is presumably a degradation product (open arrowhead in (D)). HA-tagged Pkc53E was specifically co-precipitated as detected with anti-HA antibodies (arrow). 10% of the protein extract (PE) used for the IP-Trap was loaded for comparison. BC corresponds to the Trap with only agarose beads as a control. M, pre-stained protein ladder; protein size is given in kDa.

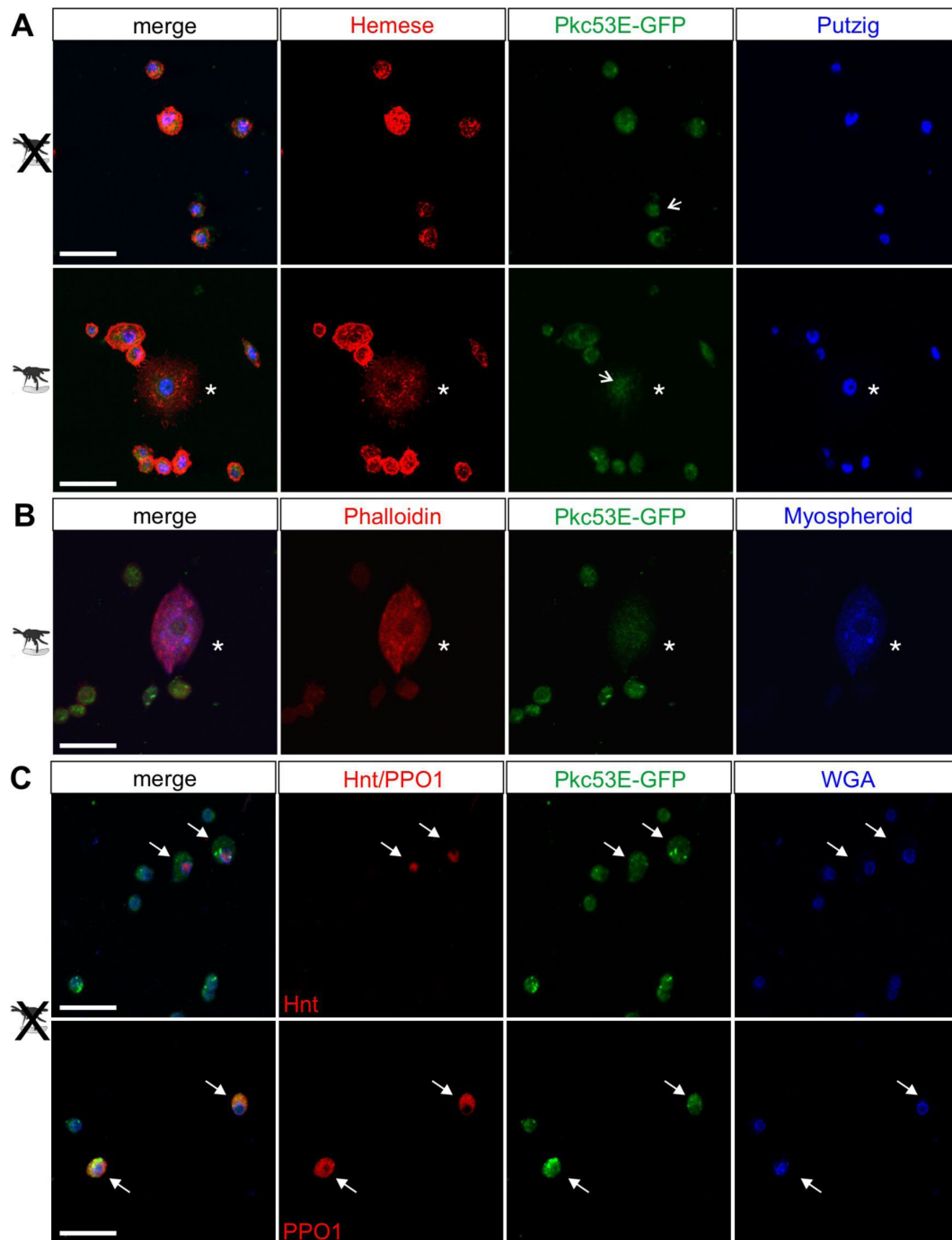


Figure 8

Pkc53E-eGFP is expressed in the cytoplasm of all hemocytes

Hemocytes derived from Pkc53E-eGFP expressing larvae, either infested or not infested with *L. boucardi* were stained with the antibodies and compounds indicated. GFP was detected directly without further antibody staining. **(A)** Pkc53E-eGFP is present in the cytoplasm of hemocytes independent of wasp infection. Note complete overlap with Hemese (red) marking all types of blood cells; Putzig (blue) labels nuclei. Asterisk denotes lamellocyte in hemolymph of infected larvae. Arrows point to nuclei expressing Pkc53E-eGFP. **(B)** Pkc53E-eGFP is expressed in the cytoplasm of a lamellocyte (asterisk), labelled with Myospheroid (blue) and Rhodamine-coupled Phalloidin (red). **(C)** Pkc53E-eGFP is enriched in the cytoplasm of crystal cells (arrow), labelled either with Hnt (red) or PPO1 (red), as indicated. Wheat germ agglutinin (WGA, blue) served to label nuclear lamina. Scale bar, 25 μ m.

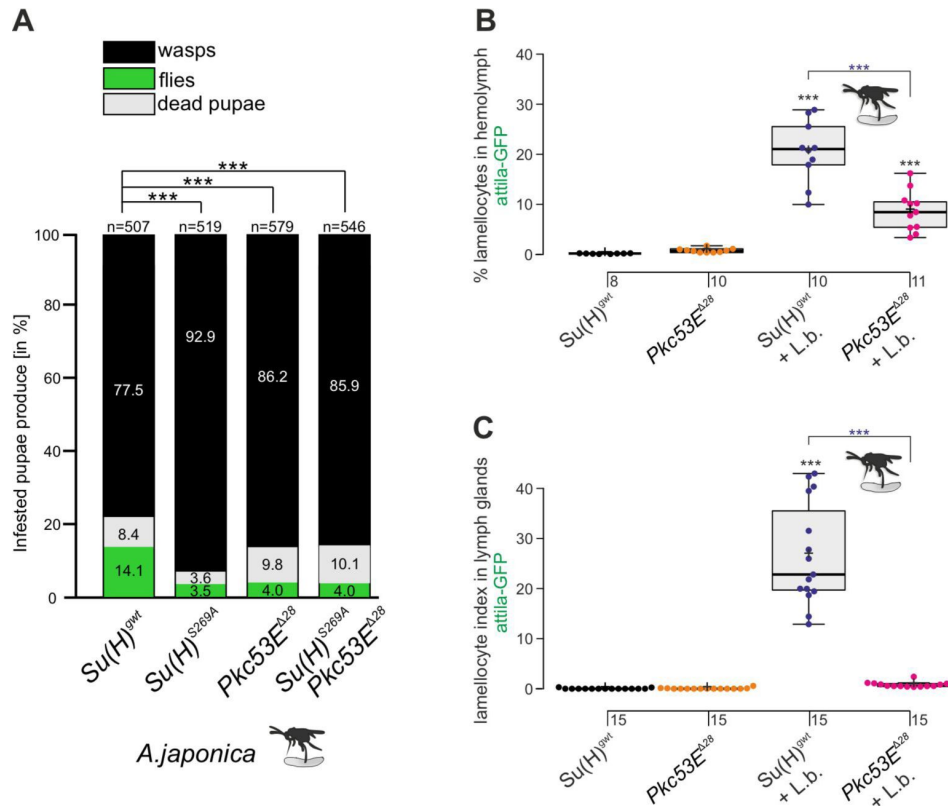


Figure 9

The *Pkc53E^{Δ28}* null mutant is immune compromised

(A) Resistance of *Pkc53E^{Δ28}* and the double mutant *Su(H)^{S269A} Pkc53E^{Δ28}*, compared to *Su(H)^{gwt}* and *Su(H)^{S269A}* for control, to the infestation with parasitic wasp strain *A. japonica*. Numbers of enclosed wasps versus flies as well as of dead pupae are presented in relation to the total of infested pupae (n, number of infested pupae). Statistical analysis by ANOVA followed by Tukey's multiple comparison test relative to control *Su(H)^{gwt}*; * $p < 0.05$; *** $p < 0.001$. **(B, C)** Quantification of lamellocytes labelled with the *attila-GFP* reporter in the circulating hemolymph (B) or in the lymph glands (C), in un-infested conditions or upon wasp infestation as indicated. Representative images of hemolymph and of lymph glands for each genotype are shown in the supplement S2. (B) Fraction of GFP-positive lamellocytes relative to the total of DAPI-stained hemocytes in the pooled hemolymph from ten larvae. Each dot represents one larval pool (n, number of experiments as shown). (C) Lamellocyte index, i.e. number of GFP-labelled cells relative to the size of the lymph gland (n=15). Statistical analysis by ANOVA followed by Tukey's multiple comparison test; *** $p < 0.001$; significant differences are colour coded.

Discussion

The Notch pathway is highly conserved between invertebrates and vertebrates, with regard to both the underlying molecular principles as well as the biological processes it is involved, including hematopoiesis and immune defence. During mammalian hematopoiesis, Notch plays a fundamental role in stem cell maintenance and proliferation as well as in the differentiation of blood cell precursors, notably in T-cell development. Accordingly, aberrant Notch signalling activity has profound consequences for blood cell homeostasis that may result in leukemia (reviewed in: Radtke et al., 2010 [↗](#); Siebel and Lendahl, 2017 [↗](#); Banerjee et al., 2019 [↗](#); Gallenstein et al., 2023 [↗](#)). Hence, the principles of the regulation of Notch signalling activity are of great interest.

Notch signals are transduced by CSL-type DNA-binding proteins that are pivotal to Notch pathway activity, as no signal transduction could occur in their absence or in the instance of a lack of DNA-binding. CSL proteins are extremely well conserved. They act as a molecular switch, either activating or repressing Notch target genes, depending on the recruited cofactors. Upon ligand binding, the Notch receptor is cleaved, and the biologically active Notch intracellular domain assembles an activator complex with CSL and further co-factors. In the absence of Notch receptor activation, however, CSL recruits co-repressors for gene silencing; in *Drosophila* repressor complex formation is mediated by Hairless (reviewed in: Maier, 2006 [↗](#); Borggreffe and Oswald, 2009 [↗](#); Kovall and Blacklow, 2010 [↗](#); Bray, 2016 [↗](#); Giaimo et al., 2021 [↗](#)). Accordingly, a loss of DNA-binding by CSL is expected to affect both, the activation of Notch targets in the presence, as well as their repression in the absence of Notch signals. This dual effect was observed the context of wing development in cells homozygous for the DNA-binding defective *Su(H)*^{S269D} variant: a failure of Notch target gene expression in areas of high Notch activity, as well as a de-repression of Notch target genes in areas outside (Frankenreiter et al., 2021 [↗](#)). Phosphorylation of Su(H) hence entails not only the inhibition of Notch activity, but likewise a de-regulation of genes silenced by Su(H)-repressor complexes. Previously, we have shown that the regulation of Notch activity during hemocyte differentiation is independent of Hairless, but rather relies on the phosphorylation of Su(H) (Frankenreiter et al., 2021 [↗](#)). Moreover, Notch activity needs to be downregulated before lamellocyte formation during parasitism, arguing for an inhibition of Notch activity rather than a de-repression of Notch target genes resulting from Su(H) phosphorylation during wasp parasitism.

Studies of *Drosophila* hematopoiesis have uncovered a pleiotropic role of Notch in all the hematopoietic compartments, where Notch activity needs to be precisely regulated to ensure blood cell homeostasis (reviewed in: Banerjee et al., 2019 [↗](#); Csordás et al., 2021 [↗](#)). During blood cell formation, Notch directs the crystal cell lineage. Accordingly, a downregulation of Notch activity causes a loss of crystal cells, whereas a gain of Notch activity results in increased numbers (Duvic et al., 2002 [↗](#); Lebestky et al., 2003 [↗](#); Terriente-Felix et al., 2013 [↗](#); Gosh et al., 2015; Frankenreiter et al., 2021 [↗](#)). In our earlier work, we have shown that the phosphorylation at S269 in the beta-trefoil domain of Su(H) impairs DNA-binding activity, and hence the capability of Su(H) to act as a transcriptional regulator in the context of blood cell development, without affecting Su(H) protein expression (Nagel et al., 2017 [↗](#); Frankenreiter et al., 2021 [↗](#)). Accordingly, phospho-deficient *Su(H)*^{S269A} mutants develop an excess of crystal cells. Now we provide evidence that Su(H) phosphorylation is likewise involved in parasitoid wasp defense, pointing to a dual use of the Notch pathway in blood cell homeostasis as well as in stress response, which is typical for the myeloid system (Banerjee et al., 2019 [↗](#)).

The primary cellular immune response of *Drosophila* to fight parasitoid wasp infestation is an encapsulation of the parasite egg to terminate its further development. Albeit metabolically extremely costly, *Drosophila* larvae have to remodel their entire hematopoietic system to generate the masses of lamellocytes required for the encapsulation (reviewed in Letournau et al., 2016; Kim-

Jo et al., 2019 [↗](#); Csordás et al., 2021 [↗](#); Hultmark & Andó, 2022 [↗](#)). There are two major sources for the lamellocytes. One is the transdifferentiation of circulating plasmatocytes, released from the sessile compartment and proliferating upon immune challenge (Márkus et al., 2009 [↗](#); Honti et al., 2010 [↗](#); Stofanko et al., 2010 [↗](#); Vanha-Aho et al., 2015 [↗](#); Anderl et al., 2016 [↗](#)). Second is a massive expansion of prohemocytes in the lymph gland followed by a differentiation to lamellocytes and their release due to the premature disintegration of the gland (Lanot et al., 2001 [↗](#); Sorrentino et al., 2002 [↗](#); Louradour et al., 2017 [↗](#); Cho et al., 2020 [↗](#)). Wasp infestation hence provokes a biased commitment to the lamellocyte lineage at the expense of crystal cells, as the two lineages are mutually exclusive (Krzemien et al., 2010 [↗](#); Tattikota et al., 2020 [↗](#); Cho et al., 2020 [↗](#); Cattenoz et al., 2020 [↗](#)).

The processes underlying the *Drosophila* immune response to wasp parasitism are well-understood, sharing many molecular details with the inflammatory responses of vertebrates (reviewed in Letourneau et al., 2016 [↗](#); Kim-Jo et al., 2019 [↗](#); Banerjee et al., 2019 [↗](#); Kharrat et al., 2022 [↗](#)). Interestingly, sterile injury of *Drosophila* larvae initiates a likewise immune response covering all aspects of wasp-mediated immune challenge (Evans et al., 2022 [↗](#)). The epidermal penetration by the wasp ovipositor causes a burst of hydrogen peroxide at the injury site via the activation of the NADPH oxidase DUOX. The oxidative stress then induces a systemic activation of Toll/NF-κB and JNK-signalling within circulating hemocytes as well as within the cells of the posterior signalling center of the lymph gland. Following cytokine release, JAK/STAT and EGFR signalling pathways are activated non-cell autonomously driving the (trans-)differentiation of (pro-)hemocytes to lamellocyte fate and lymph gland dispersal (Nappi et al., 1995 [↗](#); Schlenke et al., 2007 [↗](#); Sinenko et al., 2011 [↗](#); Gueguen et al., 2013 [↗](#); Razzell et al., 2013 [↗](#); Louradour et al., 2017 [↗](#); Chakrabarti and Visweswariah, 2020 [↗](#); Evans et al., 2022 [↗](#); reviewed in Letourneau et al., 2016 [↗](#); Banerjee et al., 2019 [↗](#)). Whereas the switch to lamellocyte fate is well elaborated, less is known about the processes underlying the simultaneous suppression of crystal cell fate. Obviously, this step requires a downregulation of Notch signalling activity, as the Notch pathway is instrumental to crystal cell fate in all the hematopoietic compartments (Duvic et al., 2002 [↗](#); Lebestky et al., 2003 [↗](#); Schlenke et al., 2007 [↗](#); Small et al., 2014 [↗](#); reviewed in Banerjee et al., 2019 [↗](#)). Our work now reveals that a parasitoid wasp attack causes a phosphorylation of Su(H) on S269 as means of an efficient and reversible way to inhibit Notch activity. This idea is consistent with earlier observations, whereby wasp parasitism quenched the expression of a Su(H)-lacZ reporter (Small et al., 2014 [↗](#)). Moreover, we provide evidence that Pkc53E is involved in Su(H) phosphorylation during blood cell development as well as during parasitoid wasp-defense in *Drosophila*. Moreover, the involvement of PKCs in wasp defense is not completely unexpected given their well-documented role in the immune responses of mammals and *Drosophila* alike. For example, *Drosophila* flies ensure a healthy gut-microbiota homeostasis by modulating DUOX activity as a pathogen-specific defense line (reviewed in Kim and Lee, 2014 [↗](#)). DUOX activation and ROS production, induced by gut pathogens was shown to be mediated by Pkc53E and Ca²⁺ via phospholipase Cβ (Ha et al., 2009 [↗](#); Lee et al., 2015 [↗](#)). Sterile wounding of the *Drosophila* embryo triggers an instantaneous Ca²⁺ flash followed by hydrogen peroxide production (Razzell et al., 2013). It is conceivable that the epidermal breach by the wasp's sting similarly induces a Ca²⁺ flash that may spark Pkc53E activation as a result. However, PKCs may be activated directly by oxidative modification even in the absence of Ca²⁺. Moreover, they may promote endogenous ROS production in a positive feedback loop (reviewed in Cosentino-Gomes et al., 2012 [↗](#)), that could support the systemic response in the larval lymph gland.

The role of PKCs in the mammalian hematopoietic and immune systems is well documented. PKCs in this context act redundantly, and presumably, this holds also true for *Drosophila* where several PKCs exist. Upon activation, PKCs may translocate into the nuclear compartment, where they can directly influence gene expression programs, for example by piloting chromatin factors as substrates, pivotal for regulating immune cell differentiation (reviewed in Lim et al., 2015 [↗](#)). Whereas *Drosophila* Pkc53E appears primarily cytoplasmic, its nuclear presence in some hemocytes is consistent with acting directly on Su(H) once activated (**Figure 8A** [↗](#)).

Phosphorylation, however, may occur also at the membrane or in the cytoplasm, since Su(H) is imported into the nucleus together with its co-regulators (Wolf et al., 2019 [DOI](#)). PKCs, including PKC α , are present in CD34+ long-term hematopoietic stem cells. Moreover, specific roles in both the myeloid and the multilymphoid lineages are known. Notably, some PKC isoforms including PKC α may play a role in Notch-dependent T-cell development. Notch pathway activity is indispensable for T-cell lineage commitment of early thymic progenitors, however, is rapidly downregulated after the β -selection phase. As treatment with phorbol ester upregulates the transcription of TCR α and β chains, a role of PKC in Notch-dependent commitment of $\alpha\beta$ T-cells has been proposed (reviewed in Altman and Kong, 2016 [DOI](#); Yui and Rothenberg, 2014 [DOI](#)). During initial T-cell development, Notch1 signalling increases in intensity, however, is abruptly downregulated at the transcriptional level following β -selection. This rapid downregulation is mediated by the inhibition of the E proteins (Yashiro-Ohtani et al., 2009 [DOI](#)), however, might in addition involve the phosphorylation of CSL by PKC α . Cell fate in more mature thymocytes is then fixed by the silencing of Notch target genes through chromatin modulators (reviewed in Yui and Rothenberg, 2014 [DOI](#)).

Appropriate silencing of Notch signalling activity in the course of T-cell development is of utmost importance, as prolonged Notch activity during β -selection predisposes the T-cells to leukemic transformation. Phosphorylation of CSL proteins by PKC α kinase offers a way for a timely and reversible deactivation of Notch signals not only in *Drosophila* but also in the mammalian system. In fact, the amino acid sequences harbouring the respective Serine residue in the CSL beta-trefoil domain are completely conserved between vertebrates and invertebrates (Wilson and Kovall, 2006 [DOI](#); Nagel et al., 2017 [DOI](#)), raising the possibility of a likewise regulatory mechanism in mammalian hematopoiesis and immunity. Indeed, respective phosphorylation of the human CSL protein at the homologous position S195 was observed in human embryonic stem cells, where differentiation was induced by phorbol ester treatment (Rigbolt et al., 2011 [DOI](#)), consistent with an involvement of PKCs in this context as well. In conclusion, our work uncovers an important role for PKC-mediated downregulation of CSL activity, thereby ensuring blood cell homeostasis and an appropriate immune response to parasitoid wasp infestation in *Drosophila*. Future work may uncover, whether similar mechanisms apply to mammalian hematopoiesis and immunity as well.

Materials and methods

Key resources are listed in S1 Table

Maintenance of parasitoid wasps and infection assay of *Drosophila melanogaster*

Leptopilina boulardi (*L. boulardi*), *Leptopilina heterotoma* (*L. heterotoma*) and *Asobara japonica* (*A. japonica*) were kindly provided by B. Häußling and J. Stökl, Bayreuth, Germany (Weiss et al., 2015 [DOI](#)). Wasp species were co-cultured with wild-type *Drosophila* larvae at room temperature. To this end, about forty 3-5 days old female wasps and twenty male wasps were co-incubated with second instar larvae for 5-7 days at room temperature. Every other day, fresh drops of honey water were added to the vial plug for feeding the wasps. After two weeks, all hatched flies were discarded. Wasps emerge about 30 days after the infestation.

For the infection assays, 50-100 staged *Drosophila* late second/early third instar larvae of the respective genotype were transferred onto apple juice plates with fresh yeast paste. 30 female and 20 male wasps aged between 3-6 days were added to the larvae, allowing to infect them for 4-6 hours. Afterwards, wasps were removed and larvae were allowed to develop further in vials with normal fly food. Wasps were only used once for each infection. Only infected larvae containing

wasp egg were used for the subsequent experiments. After infestation, hemolymph or lymph glands were prepared at the time points indicated for the particular application, or the survival rate of wasps versus *Drosophila* imago was recorded.

Fly work and genetic analyses

Fly crosses were performed with 30–40 virgin females and 20 males to avoid overcrowding and stress. Combination/recombination of fly stocks was monitored by PCR-genotyping using primers listed in S2 Table. A complete list of the Kinase mutant flies tested in the ‘larval kinase screen’ is found in **Figure 3** – supplement Table 3. As reporter lines served *atilla*-GFP (BL23540) and *PPO3*-Gal4 UAS mCD8-GFP (named *PPO3::GFP*, Dudzic et al., 2015), *hml11*-Gal4 UAS-GFP (named *hml::GFP* herein, BL30142) (Sinenko et al. 2004) and *He*-Gal4 UAS-GFP (BL8700) (Zetterval et al., 2004). For Gal4/UAS based overexpression and RNAi-mediated knockdown, we used *hml*-Gal4 (BL30141), *lz*-Gal4 (Lebestky et al., 2000; obtained from M. Crozatier, Université de Toulouse, France), UAS- μ MCas9 (VDRC 340002), UAS-HA-*Pkc53E* (this study), UAS-*white*-RNAi (BL31231), UAS-*Pkc53E*-RNAi²⁷⁴⁹¹ (BL27491), UAS-*Pkc53E*-RNAi³⁴⁷¹⁶ (BL34716), UAS-*sgRNA-Pkc53E* (VDRC341127) and UAS-*N*-RNAi (BL7078). The strain *vasa- χ TC31*, *96E-attB/TM3* (Bischof et al., 2007) served for the *Su(H)*^{gwt-mCh}, *Su(H)*^{S269A} and *Su(H)*^{S269D}/CyO-GFP, (Praxenthaler et al., 2017; Frankenreiter et al., 2021). *Su(H)*^{S269A-mCh} flies were produced in this study by a C-terminal in frame fusion of mCherry to the *Su(H)*^{S269A} mutant gene followed by genomic integration of the construct via gene engineering as outlined before (Praxenthaler et al., 2017).

Generation of the UAS-HA-*Pkc53E* fly line

Pkc53E cDNA (DGRC GH03188) was PCR amplified and subcloned via *XhoI/XbaI* in a modified pBT-HA vector, harbouring three copies of an HA-Tag generated via annealed oligos cloned into *Acc65I/XhoI* to generate pBT-3xHA-*Pkc53E*. HA-*Pkc53E* was then shuttled via *Acc65I/XbaI* in likewise opened pUAST-attB vector (Bischof et al., 2007). All cloning steps were sequence verified. Primers used for cloning are included in S2 Table. Transgenic fly lines were then generated with the help of the χ TC31 integrase-based system using 96E as landing site (Bischof et al., 2007).

Analyses of *Drosophila* hematopoietic cells and tissues

Recording total hemocyte numbers

Hemocytes were visualized by GFP fluorescence. To this end, *Su(H)*^{gwt}, *Su(H)*^{S269A} and *Pkc53E*^{Δ128} alleles, respectively, were combined with *hml*-Gal4 UAS-GFP (BL30142) and *He*-Gal4 UAS-GFP (BL8700) that together label the vast majority of hemocytes (Petraki, et al., 2015). Larvae were vortexed at maximum speed with glass beads for 2 min, and the hemolymph was collected individually in 20 μ l PBS. A fourth of the hemolymph was distributed in six Pap-pen wells of about 2 mm diameter. To ease staging and to avoid overcrowding, 4 h egg collections were used, and larvae developed in batches of about 100 animals at 25°C until wandering third instar larval stage (ca. 120 h after egg laying (AEL)). In case of wasp challenge, staged early L3 larvae were infested (ca. 90 h AEL) and bled 20–32 h thereafter to determine total hemocyte numbers. Only larvae containing wasp eggs were examined. GFP-positive hemocytes were visualized by epi-fluorescence microscopy on an Axioskop II (Zeiss, Jena), pictured with an EOS 700D camera (Canon, Japan). For counting cells, we followed earlier descriptions (Petraki et al., 2015). For reproducibility, the ‘trainable Weka segmentation’ plugin was used to demarcate cells from background (Eibe et al., 2016), followed by the ‘analyse particles’ plugin of Image J. Each Pap-pen well count was taken as a technical replicate for the individual larvae to determine the average hemocyte number per μ l, expanded by 20 for the total hemocyte number per larva.

Determination of sessile larval crystal cells

Larval crystal cells were counted according to [Frankenreiter et al., 2021](#). Briefly, staged wandering third instar larvae of the respective genotype were heated to 60°C for 10-12 min. Pictures of the posterior dorsal side were taken with a Pixera camera (ES120, Optronics, Goleta, USA) mounted to a stereo-microscope (Wild M3Z, Leica, Wetzlar, Germany) with Pixera Viewfinder 2.5. Melanized crystal cells appear as black dots, and were counted in the last two larval segments with *ImageJ* 1.51 software using *Cell Counter* tool. At least twenty larvae were scored for the statistical evaluation. For the PMA/STAU-feeding experiments, 20-30 developmentally synchronized second instar larvae were selected and grown for 24 h in complete dark at 25°C on fly food with 200 µl of 1 mM Phorbol-12-myristat-13-acetat (PMA) or PMA plus 0.2 mM Staurosporine (STAU) added to the surface of the fly food. As PMA and STAU were dissolved in DMSO before further dilution, controls were exposed accordingly to 10% DMSO on the fly food. Subsequently, wandering third instar larvae were heated and analysed as above.

Visualization and quantification of lamellocytes

The lamellocyte specific reporters *atilla*-GFP or *PPO3::GFP* strains were re/combined with *Su(H)^{gwt}*, *Su(H)^{S269A}* or *Pkc53E^{Δ28}* alleles by genetic means. Late second/early third larval instars were infested by *L. bouleari* and the number of lamellocytes was determined two days later and compared with those observed in non-infested larvae. Larvae were washed thoroughly in cold PBS and dried with a tissue and teared apart. The hemolymph of 10 larvae each was collected with a 20 µl Microloader tip (Eppendorf, Hamburg, Germany) and placed on a slide with 7 µl of Vectashield mounting medium containing DAPI. GFP-positive cells, i.e. lamellocytes were counted in relation to the total number of DAPI labelled hemocytes with a Zeiss Axioskop II and a PlanNeofluar 20x objective. 8-10 independent bleedings were performed each.

Pkc53E expression in hemocytes

Pkc53E-eGFP flies are derived from a protein trap and express endogenously a respective fusion protein (Lee et al., 2018). The hemolymph of three to five third instar *Pkc53E-eGFP* larvae at around 100 h AEL, non-infested or infested at 80 h AEL overnight with *L. bouleari*, was collected as described above in 20 µl cold PBS. Hemocytes were allowed to settle for 5-10 min in 500 µl cold PBS onto a round 18 mm glass slide placed in a 12-well microtiter plate. After fixation in 1ml 4% paraformaldehyde in PBS for 15 min at room temperature, three washes with PBS plus 0.3% TritonX100 (PBX), and a pre-incubation step in 500µl 4% normal antibodies in 4% donkey serum in PBX (anti-GFP 1:100; anti-He 1:50; anti-Hnt 1:20, anti-mys 1:10, anti-PPO1 1:3; anti-Pzg 1:500, rhodamine-coupled phalloidin (1:200-1:400). Depending on the combination, staining with Wheat Germ Agglutinin (WGA, 1:200) was performed for 15 min (hnt) and 60 min (PPO1) at room temperature. Three further washing steps and a pre-incubation step as above were followed by incubation for two hours at room temperature with suitable secondary antibodies (1:200) in the dark, followed by three additional washing steps. Cells were mounted in Vectashield by placing the round coverslip upside down on a glass slide, and pictures taken with a BioRad MRC1024 coupled to Zeiss Axioskop with a PlanNeofluar 63x objective using LaserSharp software 2000 (Zeiss, Jena, Germany).

Immunostaining and documentation of larval lymph glands

Larval lymph glands were prepared 24-36 h after wasp infection and treated as described before ([Frankenreiter et al., 2021](#)). For comparison, non-infested lymph glands were prepared. Primary antibodies used for staining: mouse anti-Hnt for crystal cells (1:20) and guinea pig anti-Pzg as nuclear marker (1:500). GFP signals were monitored directly. Secondary fluorescent antibodies were from Jackson Immuno-Research Laboratories (1:250 each). Mounted tissue was documented with a Zeiss Axioskop coupled with a BioRad MRC1024 confocal microscope using LaserSharp

software 2000 (Zeiss, Jena, Germany). For statistical evaluation at least 12 primary lobes were documented and statistically analysed by using *Image J* software (Schindelin et al., 2012 [DOI](#)). Indices represent the number of cells in relation to the size/area of the tissue (in pixel) x 10000.

RNA expression analyses

RT-PCR of *Pkc53E*^{λ128} null mutants and in hemocytes

Poly(A)⁺ RNA was isolated from 50 third instar larvae (*Pkc53E*^{λ128} and *y^{1w67c23}*) using PolyAtract System Kit 1000 according to the manufacturer's protocol, followed by a 10 min DNase I treatment at 37°C. Subsequent cDNA synthesis was conducted with qScriber cDNA Synthesis Kit according to the supplier's protocol. For amplification, a *Pkc53E* primer pair overlapping the last three introns was chosen (*Pkc53E*_RT-PCR UP and *Pkc53E*_RT-PCR LP). Tubulin 56D primers (*Tub56D*_229 UP and *Tub56D*_507 LP) served as internal controls. For primers, see S2 Table. In order to monitor *Pkc53E* expression in hemocytes, hemolymph was derived from 20 third instar *Su(H)*^{gwt} larvae, and poly(A)⁺ RNA isolated using the Dynabeads micro mRNA-Kit, with an on-beads DNase I digest, otherwise following the above protocol with primer pair *Pkc53E*_RT-PCR UP and *Pkc53E*_RT-PCR LP.

Quantification of NRE-GFP transcription in hemocytes

Su(H)^{gwt} or *Su(H)*^{S269A} stocks were genetically combined with NRE-GFP (BL30728). Hemolymph was collected from 15-30 early third instar larvae of each genotype, infested with *L. bouhardi* for 6 h at 72h AEL as described above, to be compared with non-infested *Su(H)*^{gwt} control. Poly(A)⁺ RNA was isolated directly thereafter (0-6 h value) or 24 h later (24-30 h value) with Dynabeads mRNA (micro) Kit from the cells lysed in 200 µl lysis and binding buffer according to the manufacturer's protocol, followed by an on beads DNase I digest for 10 min at 37°C. After two washing steps, mRNA was eluted with 25 µl 10 mM Tris-HCl pH8. cDNA synthesis was conducted with 15 µl using the qScriber cDNA Synthesis Kit according to the supplier's protocol. Real-time qPCR was performed as described before using the Blue S'Green qPCR Kit and the MIC magnetic induction cyclers (bms, Australia), including target and no-template controls (Kober et al., 2019 [DOI](#)). Four biological replicates with two technical replicates were each conducted. Results were compared to the ubiquitously expressed genes *cyp33* and *Tbp* as reference. Primer pairs are listed in S2 table. The micPCR[®] software version 2.12.7 was used for relative quantification of the data, based on REST and taking target efficiency into account (Pfaffl et al., 2002 [DOI](#)).

Determination of kinases and kinase assays

Screening of protein kinase candidates in silico and in vitro

To search for potential candidates in silico, GPS3.0 software was used at the lowest threshold levels, including the 40 kinases with the highest difference between score and cut-off value (Xue et al., 2011 [DOI](#)). The corresponding *Drosophila* kinases were determined with the help of flybase according to (Morrison et al., 2000 [DOI](#)). For the *in vitro* screen, a 0.5kb cDNA fragment (741-1242) encoding the *Su(H)* beta-trefoil domain (codons 247-414), was PCR-amplified and cloned via *Bam*HI/ *Eco*RI into pGEX-2T vector (Smith and Johnson, 1988 [DOI](#)) for bacterial expression and purification of the BTD-GST fusion protein. Primers used for cloning are included in S2 Table. ProQinase GmbH (Freiburg, Germany) provided the 'KinaseFinder assay service'. Briefly, BTD-GST and ³³P-ATP served as substrates for 245 human Ser/Thr kinases in multi-well plates, analysed in a microplate scintillation reader. (A) Activity of each kinase was determined, (B) corrected for substrate background activity, and (C) auto-phosphorylation (kinase activity without substrate). A ratio value between phosphorylation of BTD-*Su(H)* and kinase auto-phosphorylation ≥1 (A-B/C) was considered as phosphorylating.

In vitro ADP-Glo™ kinase assay

Drosophila pBT-3xHA-*Pkc53E* was mutated to generate the pseudo-activated form *Pkc53E^{EDDD}* (A34E/T508D/T650D/S669D) stepwise by site directed mutagenesis using the Q5® Site directed Mutagenesis Kit. Primers used for mutagenesis are included in S2 Table. *Pkc53E* as well as *PKC53E^{EDDD}* were then shuttled into a modified pMAL vector (Riggs, 1994 [\[1\]](#)) where additional restriction sites for *Acc65I*, *SacII* and *XhoI* had been included in the multiple cloning site via primer annealing. The MBP-*Pkc53E* and MBP-*Pkc53E^{EDDD}* fusion proteins were bacterially expressed and purified with Amylose resin. Additionally, activated human kinase PKCα was obtained as a positive control. The PKCα pseudo-substrate PS (RFARLGSLRQKNV) (Kochs et al., 1993 [\[2\]](#)), the wild type Su(H) peptide S^{wt} (ALFNRLRSQTVSTRY) and the phospho-deficient peptide S^{SA} (ALFNRLRAQTVSTRY) were obtained (peptides & elephants, Henningsdorf, Germany).

To test kinase activity, the ADP-Glo™ Kinase Assay system was used. Kinase assay reactions were performed in 96 well plates in a volume of 25 µl in the dark. Each reaction contained 100 µM of a kinase substrate peptide, 150 ng purified kinase and 500 µM ultra-pure ATP. To stimulate kinase activity, 150 nM Phorbol 12-myristate 13-acetate (PMA) was added to some reactions. The mixture was filled up with Kinase reaction buffer (40 mM Tris-HCl, 20 mM MgCl₂, 0.1 mg/ml BSA, pH 7.4) and incubated at room temperature for 1 h in the dark. 25 µl ADP-Glo Reagent were added and incubated for 40 min to remove residual ATP. 50 µl of Kinase Detection Reagent was applied to convert ADP to ATP. The luminescent signal was measured after 45 min using GloMax® Discover Microplate Reader (Promega, Madison, USA), kindly provided by the Department of Zoology (190z), University of Hohenheim.

NanoLC-ESI-MS/MS analysis of Su(H) peptides

Nano-LC-ESI-MS/MS experiments were performed by the Mass Spectrometry Unit at the Core Facility Hohenheim (640) on an Ultimate 2000 RSLCnano system coupled to a Nanospray Flex Ion Source and a Q-Exactive HF-X mass spectrometer (Thermo Fisher Scientific, Waltham, USA). Peptides were separated with LTQ-Orbitrap XL coupled to a nano-HPLC operated under the control of XCalibur 4.1.31.9 software (Thermo Fisher Scientific, Waltham, USA). For all measurements using the Orbitrap detector, internal calibration was as described before (Olsen et al., 2005 [\[3\]](#)). MS/MS spectra were analysed using Proteome Discoverer 2.2 (ThermoFisher Scientific, Waltham, United States), verified by manual inspection of the MS/MS spectra (Voolstra et al., 2010 [\[4\]](#)).

Generation of a α-pS269 antiserum

Rabbit polyclonal p-S269 Su(H) antiserum was generated by DAVIDS Biotechnology GmbH (Regensburg, Germany) using the synthetic phospho-peptide NRLRpSQTVSTRYLHVE. Phospho-specific antibodies were enriched in a depletion-step by affinity purification against the non-phosphorylated peptide. We note a very low affinity of the purified antiserum for Su(H) protein. The antiserum detects purified Su(H)-GST fusion proteins (wild type, S269A as well as S269D) with low affinity in Western blots, but neither native Su(H) in tissue nor hemocytes nor in Western blots, except if phosphorylated and enriched by Trap-technology.

Immunoprecipitation of mCherry and Myc-tagged Su(H)

400 adult heads or 25 larvae of each genotype were homogenized on ice in 220 µl buffer 1 (150 mM NaCl, 1% Triton X-100, 50 mM Tris-HCl pH 7.5, 0.1% SDS, supplemented with protease inhibitor cocktail) and incubated for 15 min. After a short spin, 20 µl of the supernatant was set aside as input fraction ('protein extract'). The residual supernatant was diluted with 300 µl wash buffer I (see buffer I, but without Triton X-100). 15 µl of equilibrated magnetic RFP-Trap Magnetic Agarose beads were added, incubated for 1 h at 8°C and washed three times with wash buffer I. mCherry-

trapped proteins were resolved on SDS-PAGE; Western blots were probed with rabbit anti-mCherry (1:1000) and rat anti-HA (1:500). Goat secondary antibodies coupled with alkaline phosphatase (1:1000) were used for detection.

HeLa cell culture experiments and Luciferase assays

Generation of HSV-TK 2xMyc-Su(H)-VP16

Two myc-tags were added to *Su(H)* cDNA in pBT (Maier et al., 2011 [\[1\]](#)) by insertion of the two annealed oligonucleotides Myc-Tag UP and Myc-tag LP into the *EcoRI* site (for primers, see S2 Table). The construct was subsequently shuttled via *EcoRI/XhoI* into pCDNA3.1. A VP16 activator domain was then cloned in frame at the C-terminus of 2xMyc-Su(H) by replacing the 829 bp *BspEI/ApaI* fragment of pCDNA3 2xMyc-Su(H) with a respective 1060 bp fragment of the pUAST *Su(H)-VP16* construct (Cooper et al., 2000 [\[2\]](#)). Then the CMV Promoter of pCDNA3 2xMyc-Su(H)-VP16 was replaced by the HSV-TK Promoter of the pRL TK Vector. To this end, the 1023 bp *BglIII/NheI* fragment from pRL TK was cloned into the *BglIII/SpeI* opened pCDNA3 2xMyc-Su(H)^{VP16} construct.

Transfection of HeLa cells and reporter assay

RBPj^{KO} HeLa cells were cultivated and transfected as described (Wolf et al., 2019 [\[3\]](#)). The following constructs were used: pGL3 NRE-reporter (Bray et al., 2005 [\[4\]](#)), pCDNA3 HSV-TK seeded in each well of a 12-well cell culture plate. After 24 h the cells were transfected with 500 ng pGL3 NRE-reporter, 460 ng pCDNA3 HSV-TK 2xMyc Su(H)^{VP16} and 40 ng pRL-TK. 4 h after the transfection the cells were treated with 162 nM PMA, 162 nM PMA plus 21.4 nM Staurosporine (STAU) or 21.4 nM STAU alone. Control cells were treated with the same volume of DMSO present in the other treatments. 14 hours later, cells were washed twice in PBS pH 7.4 and lysed in 75 µl 1x Passive Lysis Buffer. The Dual-Luciferase® Reporter Assay was performed according to the manufacturer's instructions.

Su(H)-VP16 expression levels were detected with anti-Myc mAB (1:500) and anti beta-tubulin as loading control (1:500). To this end, 5 x 10⁵ HeLa *RBPj*^{KO} cells were transfected and treated as above, harvested and lysed 14 h later in 300 µl binding buffer (20 mM HEPES pH 7.6, 150 mM MgCl₂, 10% glycerol, 0.05% NP-40, 1 mM DTT, ROCHE cCompleteTM ULTRA-tablet Mini protease inhibitor); 20 µl of each lysate were loaded for Western blotting.

Statistical analysis and documentation of data

Normality of the data was checked by a Shapiro-Wilk test using GraphPad Prism 9.0. In case of normally distributed data, a two-tailed analysis of variance (ANOVA) approach for multiple comparisons according to Tukey-Kramer's Honestly Significance Difference or an unpaired t-test was applied, and in the other cases the non-parametric Kruskal-Wallis sum test or Dunn's test for multiple comparisons. Quantification of RT-PCR data is based on REST, and was performed with the micPCR[®] software version 2.12.7, taking target efficiency into account (Pfaffl et al., 2002 [\[5\]](#)). In the figures, p-values are presented as ***, p<0.001; **, p<0.01; *, p<0.05; not significant, p≥0.5; the exact p-values are given in the respective source data. Pictures were assembled using ImageJ, PhotoPaint, CorelDraw and BoxPlotR software. In the box plots made by BoxPlotR, center lines show the medians; box limits indicate the 25th and 75th percentiles as determined by R software; whiskers extend 1.5 times the interquartile range from the 25th and 75th percentiles, outliers are represented by dots. Number of sample points is given in the figure or legend.

Data availability

All data are contained within the manuscript.

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Author contributions

Sebastian Deichsel, Data curation, Formal analysis, Methodology, Investigation, Resources, Writing—review and editing, Validation, Visualization; Lisa Frankenreiter, Formal analysis, Methodology, Investigation, Resources, Writing—review and editing, Validation; Johannes Fechner, Formal analysis, Methodology, Investigation, Resources, Writing—review and editing, Validation; Bernd M. Gahr, Data curation, Methodology, Investigation, Resources, Writing—review and editing, Visualization; Mirjam Zimmermann, Formal analysis, Investigation, Resources, Writing—review and editing; Helena Mastel, Investigation, Writing—review and editing; Irina Preis, Investigation, Writing—review and editing; Anette Preiss, Formal analysis, Writing—original draft, Writing—review and editing, Visualization; Anja C. Nagel, Conceptualization, Data curation, Formal analysis, Methodology, Writing—original draft, Writing—review and editing, Validation, Visualization, Supervision, Funding acquisition, Project administration.

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Reviewer #1 (Public review):

The authors previously showed in cell culture that Su(H), the transcription factor mediating Notch pathway activity in *Drosophila*, was phosphorylated on S269 and they found that a phospho-deficient Su(H) allele behaves as a moderate gain of Notch activity in flies, notably during blood cell development. Since downregulation of Notch signaling is important for the production of specialized blood cell types (lamellocytes) in response to wasp parasitism, the authors hypothesized that Su(H) phosphorylation might be involved in this cellular immune response.

Consistent with their hypothesis, the authors now show that Su(H)S269A knock-in flies display a reduced response to wasp parasitism and that Su(H) is phosphorylated upon infestation. Using *in vitro* kinase assays and a genetic screen, they identify the PKC α family member Pkc53E as the putative kinase involved in Su(H) phosphorylation and they show that Pkc53E can bind Su(H). They further show that Pkc53E deficit or its knock-down in larval blood cells results in similar blood cell phenotypes as Su(H)S269A and their epistatic analyses indicate that Pkc53E acts upstream of Su(H). Finally, they show that Pkc53E mutants also display a compromised immune response to wasp parasitism.

Strengths

The manuscript is well presented and the experiments are sound, with a good combination of genetic and biochemical approaches and several clear phenotypes backing the main conclusions. Notably Su(H)S269A mutation strongly reduces lamellocyte production. Moreover, the epistatic data are convincing, notably concerning the relationship between Notch/Su(H) and Pkc53E for crystal cell production.

Even though it is not fully established, the overall model is credible and interesting. In addition, it opens further avenues of research to study the activation of Pkc in response to an immune challenge.

Weaknesses

Apparently, the hypothesis that Pkc53E is required for Su(H) phosphorylation *in vivo* could not be directly tested due to the lack of an appropriate tool (the specificity and sensitivity of the current anti-pS269 antibody was insufficient).

Also, the poor immune response of Pkc53E mutant might rather be linked to their constitutively reduced circulating blood cell number than to a deficit in Notch/Su(H) down-regulation following wasp infestation.

<https://doi.org/10.7554/eLife.89582.2.sa3>

Reviewer #2 (Public review):

The current draft by Deischel et al., describes the role of Pkc53E in the phosphorylation of Su(H) to down regulate its transcriptional activity to mount a successful immune response upon parasitic wasp-infection. Overall, I find the study interesting and relevant especially the identification of Pkc53E in phosphorylation of Su(H) is very nice. The authors have proved the central idea linking phosphorylation of Su(H) via Pkc53E to implying its modulation of Notch activity to mount a robust immune response is now well addressed in its entirety and I find the paper indeed very interesting.

Comments on revised version:

The authors have addressed all pending concerns and I have no further comments. I indeed complement the authors for their wonderful piece of work.

Reviewer #3 (Public review):

Diechsel et al. provide important and valuable insights into how Notch signaling is shut down in response to parasitic wasp infestation in order to suppress crystal cell fate and favor lamellocyte production. The study shows that CSL transcription factor Su(H) is phosphorylated at S269A in response to parasitic wasp infestation and this inhibitory phosphorylation is critical for shutting down Notch. The authors go on to perform a screen for kinases responsible for this phosphorylation and have identified Pkc53E as the specific kinase acting on Su(H) at S269A. Using analysis of mutants, RNAi and biochemistry-based approaches the authors convincingly show how Pkc53E-Su(H) interaction is critical for remodeling hematopoiesis upon wasp challenge. I find the study interesting, and the data presented supports the overall conclusions made by the authors. The authors have addressed all my comments satisfactorily in the revised submission.

Strengths:

The manuscript is well presented, and the conclusions made are backed by genetic, biochemical and molecular biology-based approaches. Overall, the authors convincingly demonstrate how Pkc53E mediated phosphorylation of Su(H) shuts down Notch signaling during wasp infestation in *Drosophila*.

Weaknesses:

The exact molecular trigger for activation of Pkc53E is still uncharacterized and it would be interesting to know how Pkc53E gets activated during wasp infestation and whether Pkc53E gets activated turning down Notch in other stress induced scenarios.

The authors have addressed comments satisfactorily. Overall, I think the findings are interesting and would be useful to the field of developmental biology and immunology and address an important gap in the field. The most significant conclusion from the work is how Notch acts as a molecular switch during parasitic wasp infestation.

<https://doi.org/10.7554/eLife.89582.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

The authors previously showed in cell culture that Su(H), the transcription factor mediating Notch pathway activity, was phosphorylated on S269 and they found that a phospho-deficient Su(H) allele behaves as a moderate gain of Notch activity in flies, notably during blood cell development. Since a downregulation of Notch signaling was proposed to be important for the production of a specialized blood cell types (lamellocytes) in response to wasp parasitism, the authors hypothesized that Su(H) phosphorylation might be involved in this cellular immune response.

Consistent with their hypothesis, the authors show that Su(H)S269A knock-in flies display a reduced response to wasp parasitism and that Su(H) is phosphorylated upon infestation. Using in vitro kinase assays and a genetic screen, they identify the PKCα family member Pkc53E as the putative kinase involved in Su(H) phosphorylation and they show that Pkc53E can bind Su(H). They further show that Pkc53E deficit or its knock-down

in larval blood cells results in similar blood cell phenotypes as Su(H)S269A, including a reduced response to wasp parasitism, and their epistatic analyses indicate that Pkc53E acts upstream of Su(H).

Strengths

The manuscript is well presented and the experiments are sound, with a good combination of genetic and biochemical approaches and several clear phenotypes which back the main conclusions. Notably Su(H)S269A mutation or Pkc53E deficiency strongly reduces lamellocyte production and the epistatic data are convincing.

Weaknesses

The phenotypic analysis of larval blood cells remains rather superficial. Looking at melanized cells is a crude surrogate to quantify crystal cell numbers as it is biased toward sessile cells (with specific location) and does not bring information concerning the percentage of blood cells differentiated along this lineage.

In Su(H)S269A knock-in or Pkc53E zygotic mutants, the increase in crystal cells in uninfected conditions and the decreased capacity to induce lamellocytes following infection could have many origins which are not investigated. For instance, premature blood cell differentiation could promote crystal cell differentiation and reduce the pool of lamellocytes progenitors. These mutations could also affect the development and function of the posterior signaling center in the lymph gland, which plays a key role in lamellocyte induction.

Similarly, the mild decrease on resistance to wasp infestation (Fig. 2A) could reflect a constitutive reduction in blood cell numbers in Su(H)S269A larvae rather than a defective down-regulation of Notch activity.

We fully agree with the reviewer that sessile crystal cells counts are a coarse approach to capture hemocytes. However, they allowed the screening of numerous genotypes in the course of our kinase candidate screen. We recorded the hemocyte numbers in the various genetic backgrounds and with regard to wasp infestation. There was no significant difference between Su(H)S269A and Su(H)gwt control, independent of infection. This is in agreement with earlier observations of unchanged plasmatocyte numbers in *N* or *Su(H)* mutants compared to the wild type (Duvic et al., 2002). We noted, however, a small drop in hemocyte numbers in Su(H)S269D and a strong one in Pkc53ED28 mutants in both conditions relative to control. Presumably, Pkc53E has a more general role in blood cell development, which we have not further analysed. The results were included in new Figure 1_S1 and Figure 9_S1 supplements. Based on the link between hemocyte numbers and wasp resistance (e.g. McGonigle et al., 2017), we cannot exclude that the lowered resistance of Pkc53ED28 mutants regarding wasp attacks is partly due to reduced hemocyte numbers, albeit we did not see significant differences between either Su(H)S269A, nor Pkc53ED28 nor the double mutant. We have included this notion in the text.

Lamellocytes arise in response to external challenges like parasitoid wasp infestation by trans-differentiation from larval plasmatocytes, and by maturation of lamellocyte precursors in the lymph gland, yet barely in the Su(H)S269A and Pkc53ED28 mutants.

We find it hard to envisage, however, that a premature differentiation of plasmatocytes into crystal cells in our case could deplete the pool of lamellocyte progenitors in the hemolymph. (Is there a precedent?). Crystal cells make up about 5% of the hemocyte pool; they are increased max. 2 fold in the Su(H)S269A and Pkc53E mutants. Even if these extra crystal cells (now ~10%) had arisen by premature differentiation, there should be still enough plasmatocytes (~80%) remaining with a potential to further divide and transdifferentiate into lamellocytes.

Indeed, we cannot exclude an effect of the Su(H)S269A mutant on the development and function of the posterior signaling center of the lymph gland. We noted, however, a slight but significant enlargement of the PS in the Su(H)S269A mutant, that to our understanding cannot explain the reduced lamellocyte numbers.

Whereas the authors also present targeted-knock down/inhibition of Pkc53E suggesting that this enzyme is required in blood cells to control crystal cell fate (Fig. 6), it is somehow misleading to use lz-GAL4 as a driver in the lymph gland and hml-GAL4 in circulating hemocytes as these two drivers do not target the same blood cell populations/steps in the crystal cell development process.

We fully agree with the reviewer that the two driver lines target different blood cell populations/ steps in hematopoiesis. The *hml*-Gal4 driver is regarded pan-hemocyte, common to both plasmatocytes and pre-crystal cells (e.g. Tattikota et al., 2020). It has been reported to drive specifically within differentiated hemocytes prior to or at the stage of crystal cells commitment (Mukherjee et al., 2011). Hence, *hml*-Gal4 appeared suitable to hit sessile and circulating hemocytes prior to final differentiation into crystal cells or lamellocytes, respectively.

In the lymph gland, however, *hml* is expressed within the cortical zone, where it appears specific to the plasmatocytes lineage, and not present in the crystal cell precursors (Blanco-Obregon et al., 2020). In contrast, *lz*-Gal4 is specific to the differentiating crystal cells in both lineages, i.e. in circulating and sessile hemocytes and in the lymph gland. Hence, we choose *lz*-Gal4 instead of *hml*-Gal4 at the risk of driving markedly later in the course of crystal cell differentiation. We included the reasoning in the text. Overall, we feel that this choice does not limit our conclusions.

In addition, the authors do not present evidence that Pkc53E function (and Su(H) phosphorylation) is required specifically in blood cells to promote lamellocyte production in response to infestation.

We have tried to address this interesting question by several means. Firstly, we show that Pkc53E is indeed expressed in the various cell types of larval hemocytes, shown in a new Figure 8 and Figure 8_S1 supplement. I.e., there is the potential of Pkc53E to promote lamellocyte formation. Moreover, RNAi-mediated downregulation of Pkc53E within hemocytes affected crystal cell formation similar to the Pkc53ED28 mutant, in agreement with a specific requirement within blood cells (Figure 6). Finally, we show a major drop in Notch target gene transcription (NRE-GFP) in response to wasp infestation within isolated hemocytes from Su(H)gwt in contrast to Su(H)S269A larvae (see new Figure 1 G). These data show that Su(H)-mediated Notch activity must be downregulated in hemocytes prior to lamellocyte formation in agreement with our hypothesis.

Finally, the conclusion that Pkc53E is (directly) responsible for Su(H) phosphorylation needs to be strengthened. Most importantly, the authors do not demonstrate that Pkc53E is required for Su(H) phosphorylation in vivo (i.e. that Su(H) is not phosphorylated in the absence of Pkc53E following infestation).

We would very much like to show respective results. Unfortunately, the low affinity of our pS269 antibody does not allow any in situ or in vivo experiments. We very much hope to obtain a more specific phosphoS269-Su(H) antibody allowing us further in situ studies, and show, for example co-localization with Pkc53E.

In addition, the in vitro kinase assays with bacterially purified Pkc53E (in the presence of PMA or using an activated variant of Pkc53E) only reveal a weak activity on a Su(H)

| peptide encompassing S269 (Fig. 4).

The reviewer correctly notes the poor activity of our purified Pkc53EEDDD kinase. This low activity also holds true for the standard peptide (PS), which in fact is even less well accepted than the Swt substrate. Indeed, the commercially available PKCα is a magnitude more active. Whether this reflects the poor quality of our isolated protein compared to the commercial PKCα, or whether it reflects a true biochemical property of Pkc53E remains to be shown in the future. We noted this observation in the manuscript.

Moreover, while the authors show a coIP between an overexpressed Pkc53E and endogenous Su(H) (Fig. 7) (in the absence of infestation), it has recently been reported that Pkc53E is a cytoplasmic protein in the eye (Shieh et al. 2023), calling for a direct assessment of Pkc53E expression and localization in larval blood cells under normal conditions and upon infestation.

Indeed, it is interesting that a Pkc53E-GFP fusion protein is cytoplasmic in the eye. The construct reported by Shieh et al. however, i.e. the B-isoform, is preferentially expressed in photoreceptors, where it regulates the de-polymerization of the actin cytoskeleton.

Due to the eye-specific expression, we unfortunately cannot use the Pkc53E-B-GFP construct to test for Pkc53E's distribution in other tissues.

As this construct is of little use for studying hematopoiesis, we have instead used Pkc53E-GFP (BL59413) derived from a protein trap: again, GFP is primarily seen in the cytoplasm of hemocytes, including lamellocytes of infected larvae. However, in a small number of hemocytes, GFP appears to be also nuclear (Fig. 8A), leaving the possibility that activated Pkc53E may localize to the nucleus, eventually phosphorylating Su(H) and downregulating Notch activity. As Su(H) enters the nucleus piggy-back with NICD, however, phosphorylation may as well occur at the membrane or within the cytoplasm. We note, however, that these hypotheses require a much more detailed analysis.

Furthermore, the effect of the PKCα agonist PMA on Su(H)-induced reporter gene expression in cell culture and crystal cell number in vivo is somehow consistent with the authors hypothesis, but some controls are missing (notably western blots to show that PMA/Staurosporine treatment does not affect Su(H)-VP16 level) and it is unclear why STAU treatment alone promotes Su(H)-VP16 activity (in their previous reports, the authors found no difference between Su(H)S269A-VP16 and Su(H)-VP16) or why PMA treatment still has a strong impact on crystal cell number in Su(H)S269A larvae.

We have added a Western blot showing that the treatment does not affect Su(H)-VP16 expression levels (Figure 5_supplement 1). As STAU is a general kinase inhibitor, it may obviate any inhibitory phosphorylation of Su(H)-VP16 in the HeLa cells, e.g. that by Akt1, CAMK2D or S6K which pilot T271, phosphorylation of which is expected to affect the DNA-binding of Su(H) as well (Figure 3_supplement 2). Moreover, in the previous report, we used different constructs with regard to the promoter, and we used RBPJ instead of Su(H), which may explain some of the discrepancies. As PMA is not specific to just Pkc53E, the altered crystal cell numbers may result from the influence on other kinases involved in blood cell homeostasis, as predicted by our genetic screen (Figure 3_supplement 1).

Reviewer #1 (Recommendations For The Authors):

(1) The authors should provide a more elaborate examination of larval blood cell types and blood cell counts under normal conditions and following infestation in the different zygotic mutants as well as upon Pkc53 knock-down. A thorough examination of PSC integrity should be performed and the maintenance of core blood cell progenitors

examined. The authors should also clarify when after infestation the LG and larval bleeds are analyzed.

- a more elaborate examination of larval blood cell types:

- examination of larval blood cell counts under normal conditions: hemocyte # in gwt, SA, SD, & Pkc

- examination of larval blood cell counts after infestation: hemocyte # in gwt, SA, SD, & Pkc

- thorough examination of PSC integrity: in gwt, SA, SD, & Pkc

- thorough examination of blood cell progenitors: in gwt, SA, SD, & Pkc

- clarify timing

Hemocyte numbers of the various genotypes and conditions were recorded and are presented in Figure 1_S1 and Figure 9_S1. Timing was elaborated in the text and the Methods section.

(2) The authors should clarify why they use lz-GAL4 or hml-GAL4 and what we can infer from using these different drivers.

See above. The reasoning was included in the text.

(3) The percentage of hatching of Su(H)S269A and Su(H)gwt flies in the absence of infestation should also be scored; a small decrease in Su(H)S269A viability might explain the observed differences in survival to wasp infestation. Absolute blood cell numbers (in the absence of infestation) have also been correlated with survival to infection and should be checked.

Percentage of the emerging flies and hemocyte numbers in the absence of infestation were recorded and included in Figure 2, Figure 1_S1, Figure 9_S1.

(4) Whereas the impact of Su(H)S269A or Pkc53E mutation on lamellocytes production is clear, there is still a substantial reduction in crystal cell production following infestation. So I wouldn't conclude that the Su(H) larvae are "unable" to detect this immune challenge or respond to it (line 116).

Thank you for the hint, we corrected the text.

(5) The expression and localization of Pkc53E in larval blood cells should be investigated, for instance using the Pkc53E-GFP line recently published by Shieh et al. (or at least at the RNA level).

Firstly, we confirmed expression of Pkc53E in hemocytes by RT-PCR (Figure 8_S1 supplement). Secondly, expression of Pkc53E-GFP was monitored in hemocytes (Figure 8). To this end, we used the protein trap (BL59413), since the one published by Shieh et al., 2023 is restricted to photoreceptors.

(6) It would be interesting to test the anti-pS269 antibody in immunostaining (using Su(H)S269A as negative control).

Unfortunately, the pS269 antiserum does not work in situ at all.

(7) *The authors must perform a western blot with anti-pS269 in Pkc53e mutant to show that Su(H) is not phosphorylated anymore after wasp infestation.*

The blot gives a negative result.

(8) *It is surprising that no signal is seen in the absence of infestation with anti-pS269: the fact that Su(H)S269A have more crystal cells suggest that there is a constitutive level of phosphorylation of Su(H).*

We fully agree: In the ideal world, we would expect a low level of S269 phosphorylation in the wild type as well. However, given the lousy specificity of our antibody, we were happy to see phospho-Su(H) in infected larvae. We are currently working hard to get a better antibody.

(9) *The authors should check Su(H)-VP16 levels and phosphorylation status after PMA and/or staurosporine treatment. Some clarifications are also needed to explain the impact of PMA in Su(H)S269 larvae (this clearly suggests that PKC has other substrates implicated in crystal cell development).*

Su(H)-VP16 expression levels were monitored by Western blot and were not altered conspicuously (Figure 5_1 supplement). Presumably, Pkc53E is not the only kinase involved in Su(H) phosphorylation or the transduction of stress signals. Moreover, PMA may have a more general effect on larval development and hematopoiesis affecting both genotypes. We included this reasoning in the text.

(10) *Concerning the redaction, the authors forgot to mention and discuss the work of Cattenoz et al. (EMBO J 2020). The presentation of the screen for kinase candidates could be streamlined and better illustrated (notably supplement table 4, which would be easier to grasp as a figure/graph). The discussion could be shortened (notably the part on T cells), and I don't really understand lines 374-376 (why is it consistent?).*

We are sorry for omitting Cattenoz et al. 2020, which we have now included. We fully agree that this paper is of utmost importance to our work. We streamlined the screen and included a new figure in addition to table 4 summarizing the results graphically (Figure 3_S1 supplement). We cut on the T cell part and omitted the strange lines.

Reviewer #2 (Public Review):

Summary:

The current draft by Deischel et.al., entitled "Inhibition of Notch activity by phosphorylation of CSL in response to parasitization in Drosophila" describes the role of Pkc53E in the phosphorylation of Su(H) to downregulate its transcriptional activity to mount a successful immune response upon parasitic wasp-infection. Overall, I find the study interesting and relevant especially the identification of Pkc53E in phosphorylation of Su(H) is very nice. However, I have a number of concerns with the manuscript which are central to the idea that link the phosphorylation of Su(H) via Pkc53E to implying its modulation of Notch activity. I enlist them one by one subsequently.

Strengths:

I find the study interesting and relevant especially because of the following:

(1) *The identification of Pkc53E in phosphorylation of Su(H) is very interesting.*

(2) The role of this interaction in modulating Notch signaling and thereafter its requirement in mounting a strong immune response to wasp infection is also another strong highlight of this study.

Weaknesses:

(1) Epistatic interaction with Notch is needed: In the entire draft, the authors claim Pkc53E role in the phosphorylation of Su(H) is down-stream of notch activity. Given the paper title also invokes Notch, I would suggest authors show this in a direct epistatic interaction using a Notch condition. If loss of Notch function makes many more lamellocytes and GOF makes less, then would modulating Pkc53E (and SuH)) in this manifest any change? In homeostasis as well, given gain of Notch function leads to increased crystal cells the same genetic combinations in homeostasis will be nice to see.

While I understand that Su(H) functions downstream of Notch, but it is now increasingly evident that Su(H) also functions independent of Notch. An epistatic relationship between Notch and Pkc will clarify if this phosphorylation event of Su(H) via Pkc is part of the canonical interaction being proposed in the manuscript and not a non-canonical/Notch pathway independent role of Su(H).

This is important, as I worry that in the current state, while the data are all discussed in light of Notch activity, any direct data to show this affirmatively is missing. In our hands we do find Notch independent Su(H) function in immune cells, hence this is a suggestion that stems from our own personal experience.

The role of Notch in *Drosophila* hematopoiesis, notably during crystal cell development in both hematopoietic compartments is well established; likewise the role of Su(H) as integral signal transducer in this context (e.g. Duvic et al., 2002). Not only promotes Notch activity crystal cell fate by upregulating target genes, at the same time it prevents adopting the alternative plasmatocyte fate (e.g. Terriente-Felix et al., 2013). We could confirm the downregulation of Notch target gene expression in response to wasp infestation by qRT-PCR, which was discovered earlier by Small et al. (2014). This is clearly in favor of a repression of Notch activity rather than a relief of inhibition by Su(H). A ligand-independent activation of Notch signaling has been uncovered in the context of crystal cell maintenance in the lymph gland involving Sima/Hif- α , including Su(H) as transcriptional mediator (Mukherjee et al., 2011). However, we are unaware of a respective Su(H) activity independent of Notch.

Certainly, Su(H) acts independently of Notch in terms of gene repression. Here, Su(H) forms a repressor complex together with H and co-repressors Groucho and CtBP to silence Notch target genes. Accordingly, loss of Su(H) or H may induce the upregulation of respective gene expression independent of Notch activity. This has been demonstrated, for example, during wing and heart development (Klein et al., 2000; Kölzer, Klein, 2006; Panta et al., 2020). Moreover, during axis formation of the early embryo, global repression is brought about by Su(H) and relieved by activated Notch (Koromila, Stathopoulous, 2019). In all these instances, Su(H) is thought to act as a molecular switch, and the activation of Notch causes a strong expression of the respective genes. Likewise, the loss of DNA-binding resulting from the phosphorylation of Su(H) allows the upregulation of repressed Notch target genes in wing imaginal discs, e.g. *dpm*, as we have demonstrated before with overexpression and clonal analyses (Nagel et al. 2017; Frankenreiter et al., 2021). However, H does not contribute to crystal cell homeostasis, i.e. de-repression of Notch target genes does not appear to be a major driver in this context, asking for additional mechanisms to downregulate Notch activity. Our work provides evidence that these inhibitory mechanisms involves the phosphorylation of Su(H) by Pkc53E. Formally, we cannot exclude alternative mechanisms. Hence, we have tried to avoid the direct link between Su(H) phosphorylation and the inhibition of Notch activity throughout the text, including the title. Moreover, we have discussed the possible

consequences of Su(H) lack of DNA binding, interfering either with the activation of Notch target genes or abrogating their repression.

In addition, we have performed new experiments addressing the epistasis between Notch and Su(H) during crystal cell formation (Figure 1_supplement 1). To this end, we knocked down Notch activity in hemocytes by RNAi (hml::N-RNAi) in the Su(H)gwt and Su(H)S269A background, respectively. Indeed, Notch downregulation strongly impairs crystal cell development independent of the genetic background as expected if Notch were epistatic to Su(H). We attribute the slightly elevated crystal cell numbers observed in the Su(H)S269A background to the increase in the embryonic precursors (see Fig. 4; Frankenreiter et al. 2021). Of note, the Notch gain of function allele Ncos479 also displayed a likewise increase in embryonic crystal cell precursors as well as in crystal cells within the lymph gland (Frankenreiter et al. 2021).

(2) Temporal regulation of Notch activity in response to wasp-infection and its overlapping dynamics of Su(H) phosphorylation via Pkc is needed:

First, I suggest the authors to show how Notch activity post infection in a time course dependent manner is altered. A RT-PCR profile of Notch target genes in hemocytes from infected animals at 6, 12, 24, 48 HPI, to gauge an understanding of dynamics in Notch activity will set the tone for when and how it is being modulated. In parallel, this response in phospho mutant of Su(H) will be good to see and will support the requirement for phosphorylation of Su(H) to manifest a strong immune response.

Indeed, it would be extremely nice to follow the entire processes in every detail, ideally at the cellular level. The challenge, however, is quantities. The mRNA isolated from hemocytes could be barely quantified, although the subsequent ct-values were ok. We quantified NRE-GFP expression, introduced into Su(H)gwt and Su(H)S269A, as well as atilla expression. We were able to generate data for two time slots, 0-6 h and 24-30 h post infection. The data are provided in the extended Figure 1G, and show a strong drop of NRE-GFP in the infected Su(H)gwt control compared to the uninfected animals, whereas expression in Su(H)S269A plateaus at around 60%-70% of the infected Su(H)gwt control. Atilla expression jumps up in the control, but stays low in Su(H)S269A hemocytes.

Second, is the dynamics of phosphorylation in a time course experiment is missing. While the increased phosphorylation of Su(H) in response to wasp-infestation shown in Fig.2B is using whole animal, this implies a global down-regulation of Su(H)/Notch activity. The authors need to show this response specifically in immune cells. The reader is left to the assumption that this is also true in immune cells. Given the authors have a good antibody, characterizing this same in circulating immune cells in response to infection will be needed. A time course of the phosphorylation state at 6, 12, 24, 48 HPI, to gauge an understanding of this dynamics is needed.

We really would love to do these experiments. Unfortunately, our pS269 antibody is rather lousy. It does not allow to detect Su(H) protein in tissue or cells, nor does it work on protein extracts in Westerns or for IP. Hence, we have no way so far to demonstrate cell or tissue specificity of Su(H) phosphorylation. So far, we were lucky to detect mCherry-tagged Su(H) proteins pulled down in rather large amounts with the highly specific nano-bodies. We have tried very hard to repeat the experiment with hemolymph and lymph glands only, but we have failed so far. Hence, we have to state that our antibody is neither suitable for in vivo analyses, nor for a detection of phospho-Su(H) at lower levels.

The authors suggest, this mechanism may be a quick way to down-regulate Notch, hence a side by side comparison of the dynamics of Notch down-regulation (such as by doing

RT-PCR of Notch target genes following different time point post infection) alongside the levels of pS269 will strengthen the central point being proposed.

We fully agree and hope to address these issues in the future by improving our tools.

Last, in Fig7. the authors show Co-immuno-precipitation of Pkc53EHA with Su(H)gwt-mCh 994 protein from Hml-gal4 hemocytes. I understand this is in homeostasis but since this interaction is proposed to be sensitive to infection, then a Co-IP of the two in immune cells, upon infection should be incorporated to strengthen their point.

We do not fully agree with the reviewer. Although we also think that the interaction between Pkc53E and Su(H) might occur more frequently upon infection, we propose that this is a transient process occurring in several but not all hemocytes at a given time. Moreover, in the described experiment, Pkc53E-HA was expressed in hemocytes via the UAS/Gal4 system. We cannot exclude that this approach causes an overexpression. Hence, we would not expect considerable differences between unchallenged and infested animals.

(3) In Fig 5B, the authors show the change in crystal cell numbers as read out of PMA induced activation of Pkc53E and subsequent inhibition of Su(H) transcriptional activity, I would suggest the authors use more direct measures of this read out. RT-PCR of Su(H) target genes, in circulating immune cells, will strengthen this point. Formation of crystal cells is not just limited to Notch, I am not convinced that this treatment or the conditions have other affect on immune cells, such as any impact on Hif expression may also lead to lowering of CC numbers. Hence, the authors need to strengthen this point by showing that effects are direct to Notch and Su(H) and not non-specific to any other pathway also shown to be important for CC development.

We agree with the Reviewer that the rather general influence of PMA on PKCs might present a systemic stress to the animal. For example, we observed a slight drop of crystal cell numbers also in Su(H)S269A, suggesting other kinases apart from Pkc53E were affected that are involved in crystal cell homeostasis. We have included this notion in the text. To provide more conclusive evidence we also fed Staurosporine to the larvae which reversed the PMA effect. In addition, we assayed the expression of NRE-GFP in hemocytes of infected animals by qRT-PCR, and observed a strong drop in the infected versus uninfected control but less so in Su(H)S269A. The new data are provided in extended Figures 1G and 5B.

(4) In addition to the above mentioned points, the data needs to be strengthened to further support the main conclusions of the manuscript. I would suggest the authors present the infection response with details on the timing of the immune response. Characterization of the immune responses at respective time points (as above or at least 24 and 48 HPI, as norms in the field) will be important. Also, any change in overall cell numbers, other immune cells, plasmatocytes or CC post infection is missing and is needed to present the specificity of the impact. The addition of these will present the data with more rigor in their analysis.

Total hemocyte numbers of the various genotypes, i.e. control, Su(H)S269A, Su(H)S269D, and Pkc53ED28 were included before and after wasp infestation in supplemental Figures 1_S1 and 9_S1.

(5) Finally, what is the view of the authors on what leads to activation of Pkc53E, any upstream input is not presented. It will be good to see if wasp infection leads to increased Pkc53 kinase activity.

The analysis of the full process is an ongoing project. We propose that ROS is produced upon the wasps' sting, which is to trigger the subsequent cascade of events. These have to end with activation of Pkc53E in the presumptive pre-lamellocyte pool of both lineages, i.e. in plasmatocyte of the hemolymph, presumably in the sessile compartment (Tattikotta et al., 2021) and at the same time in the lymph gland cortex harboring the LM precursors (Blanco-Obregon et al., 2020). One of the known upstream kinases, Pdk1 has a similar impact on crystal cell development as Pkc53E, making its involvement likely. Moreover, we think that other PKCs influence the process as well.

Without a good read out, e.g. a functional pSu(H) antiserum working in situ or a Pkc-activity reporter, it will be quite difficult to follow up this question. However, we already know that Pkc53E is expressed in hemocytes of all types independent of wasp infestation, in agreement with a role during lamellocyte differentiation. We hope to unravel the process in more of it in the future.

Overall, I think the findings in the current state are interesting and fill an important gap, but the authors will need to strengthen the point with more detailed analysis that includes generating new data and also presenting the current data with more rigor in their approach. The data have to showcase the relationship with Notch pathway modulation upon phosphorylation of CSL in a much more comprehensive way, both in homeostasis and in response to infection which is entirely missing in the current draft.

Reviewer #3 (Public Review):

Diechsel et al. provide important and valuable insights into how Notch signalling is shut down in response to parasitic wasp infestation in order to suppress crystal cell fate and favour lamellocyte production. The study shows that CSL transcription factor Su(H) is phosphorylated at S269A in response to parasitic wasp infestation and this inhibitory phosphorylation is critical for shutting down Notch. The authors go on to perform a screen for kinases responsible for this phosphorylation and have identified Pkc53E as the specific kinase acting on Su(H) at S269A. Using analysis of mutants, RNAi and biochemistry-based approaches the authors convincingly show how Pkc53E-Su(H) interaction is critical for remodelling hematopoiesis upon wasp challenge. The data presented supports the overall conclusions made by the authors. There are a few points below that need to be addressed by the authors to strengthen the conclusions:

(1) The authors should check melanized crystal cells in Su(H)gwt and Su(H)S269A in presence of PMA and Staurosporine?

Thank you for the suggestion. We included the results of PMA + Staurosporine feeding into an extended Fig. 5B; they match those from the HeLa cells. Unfortunately, Staurosporine alone was lethal for the larvae at various concentrations, presumably owing to the overarching inhibition of kinase activity. This global effect also explains the high crystal cell numbers in the control fed with PMA + STAU compared to the untreated animals, as the downregulation of many kinases results in higher crystal cell numbers, a fact uncovered in our genetic screen.

(2) Data for number of dead pupae, flies eclosed, wasps emerged post infestation should be monitored for the following genotypes and should be included:

Pkc53EΔ28_, Su(H)S269A_, Pkc53EΔ28 Su(H)S269A, Su(H)S269D, Su(H)S269D Pkc53EΔ28

We extended the data with and without infection. The respective data are shown in a new Fig. 9 and an extended Fig. 2, except for the Su(H)S269D allele. Su(H)S269D is larval lethal, i.e. dies too early for wasp development, and hence could not be included in the assay. Overall, Pkc53EΔ28 matched Su(H)S269A_.

(3) The exact molecular trigger for activation of Pkc53E upon wasp infestation is not clear.

Indeed, and we would love to know! Perhaps, the generation of Ca²⁺ by the wasp's breach of the larval cuticle results in Pkc53E activation. The generation of ROS could be involved as well. At this point, we can only speculate. We hope to be able in the future to obtain direct experimental evidence for the one or the other hypothesis.

(4) The authors should check if activating ROS alone or induction of Calcium pulses/DUOX activation can mimic this condition and can trigger activation of Pkc53E and thereby cause phosphorylation of Su(H) at S269

The reviewer's suggestions open up a new field of investigations, and are hence beyond of the scope of this article. However, we want to pursue the research in this direction, albeit we realize that counting crystal cells is too coarse but to give a first impression, and that lamellocytes may form already by breaching the larval cuticle. A major challenge shall be direct measurements of Pkc53E activation. To date, we have no tools for this, but ideally, we would like to have a direct, biochemical read out. Although we have been unsuccessful in the past, we want to develop a strong and specific phospho-S269 antibody that is also working in situ. Alternatively, we think of developing a PS-phosphorylation reporter, to allow reasonably addressing these questions.

(5) Does Pkc53E get activated during sterile inflammation?

We are in the process of addressing this issue, however, feel that this topic is beyond the scope of this paper. Our preliminary experiments, however, support the notion of a phospho-dependent regulation of Su(H) also in this context.

Reviewer #3 (Recommendations For The Authors):

The authors provide a graphical representation of major phenotypes that form the basis of their investigation and conclusions but have not supplemented the quantitation with images that represent these phenotypes. The authors need to include the following data to strengthen their conclusions:

(1) The authors should include representative images for each of the genotypes/conditions (in presence and absence of wasp infestation) based on which corresponding plots have been made in Figure 1. Please include this for both circulating lamellocytes in the hemolymph and in the lymph glands since this is one of the main figures presenting the key findings.

The data have been included in Figure 1-S2 supplement.

(2) Please include representative images of LG with Hnt staining and corresponding images for melanization for each of the genotypes used in the plots in Figure 6A and B.

The data have been included in Figure 6-S2 supplement.

(3) Representative images for each of the genotypes in Figure 7A & B should be included (circulating crystal cells and lymph gland crystal cell numbers).

Representative images for each of the genotypes for Fig. 7A have been included in Figure 7-S1 and for the old Fig. 7B in Figure 9-S2 supplement, respectively.

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